2-D Unstructured Compressible Navier–Stokes Solver Benchmark Report

`cns2d` — submitted solver

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Abstract

This report documents `cns2d`, an original cell-centred finite-volume solver for the two-dimensional compressible Navier–Stokes equations on mixed unstructured meshes, written in C++17 with MPI and METIS. The spatial discretization is second order through piecewise-linear inverse-distance least-squares reconstruction of the primitive variables with a Venkatakrishnan limiter; the inviscid flux is a Roe flux with a Harten–Hyman entropy fix extended to the linear waves, with HLLC and Rusanov also implemented and HLLC serving as a positivity fallback. Viscous terms use the Newtonian stress tensor and Fourier heat flux built from the same least-squares gradients. Steady cases are advanced by an implicit pseudo-time march with local time stepping and a matrix-free LU-SGS inner solve; the vortex-shedding case uses a true dual-time BDF2 scheme with an outer physical-time loop and frozen history levels. The mesh is partitioned once with METIS k-way on the cell dual graph, and iteration-time communication is strictly neighbour-scoped point-to-point halo exchange: no rank holds the full mesh or the full conservative state during iterations.

Verification is quantitative: face-closure error 1.0×10^{-16}, volume closure 1.1×10^{-10}, cell area against the boundary line integral 1.4×10^{-14}, exactness of the least-squares gradient on a linear field 3.1×10^{-13}, a discrete conservation defect of 9.2×10^{-17}, and agreement of the analytic Jacobian-vector product with central finite differences to 1.7×10^{-8} over 20 000 random states. The slip-wall flux is exactly $[0, pn_x, pn_y, 0]$, and force coefficients agree to within 5×10^{-4} relative across $np = 1, 2, 4, 8$.

All eight required cases were run with the supplied production controls, and section 10 reports each with the convergence status the solver itself recorded. Two defects found during preparation of this report are documented rather than hidden: a steady termination test that normalised residuals by the impulsive-start transient and could stop while forces were still moving (section 8.5), and a latent sign error in the unused HLLC star state that is now covered by a dedicated regression test (section 8.6). Added dissipation on the linear waves, needed to suppress a stagnation-point checkerboard mode, is disclosed and quantified in section 4.3.

Contents

1 Introduction

The benchmark asks for an original, parallel, second-order-accurate finite-volume solver for the compressible Navier–Stokes equations on the supplied unstructured CGNS meshes, and for eight cases to be run to convergence with contract-compliant output. The eight required cases are listed in table 1. They span a deliberately wide range: an essentially incompressible inviscid flow whose exact drag is zero, a transonic case with a shock on the body, a supersonic case with a detached bow shock, three laminar airfoil cases at $Re = 5000$ that add a boundary layer and skin friction, a steady separated cylinder wake at $Re = 20$, and an unsteady von Kármán vortex street at $Re = 200$ that must be integrated in physical time.

This is intended as a technical report rather than a run log. Each numerical choice is stated as it is implemented in the source, and section 10 connects those choices to what the runs actually produced — including where the results fall short of the requested targets. Section numbering follows the required report outline.

Table 1: The eight required benchmark cases with the boundary-family mapping read from each case file. Family names differ between the two meshes and are never hard-coded in the solver.

Case id	Mesh	Physics	BC mapping	Max steps
`naca0012_m015_inviscid`	NACA0012_H2	inviscid, $M_\infty = 0.15$	`bc-2`→far, `bc-4`→slip	20 000
`naca0012_m080_inviscid`	NACA0012_H2	inviscid, $M_\infty = 0.80$	`bc-2`→far, `bc-4`→slip	30 000
`naca0012_m200_inviscid`	NACA0012_H2	inviscid, $M_\infty = 2.0$	`bc-2`→far, `bc-4`→slip	40 000
`naca0012_m015_laminar_re5000`	NACA0012_H2	laminar, $M_\infty = 0.15$	`bc-4`→no-slip	40 000
`naca0012_m080_laminar_re5000`	NACA0012_H2	laminar, $M_\infty = 0.80$	`bc-4`→no-slip	40 000
`naca0012_m200_laminar_re5000`	NACA0012_H2	laminar, $M_\infty = 2.0$	`bc-4`→no-slip	50 000
`cylinder_m010_laminar_re20`	CylinderB1	laminar, $Re = 20$	`WALL`→no-slip, `FAR`→far	30 000
`cylinder_m010_laminar_re200`	CylinderB1	laminar, $Re = 200$	`WALL`→no-slip, `FAR`→far	30 000

2 Governing Equations and Nondimensionalization

The solver advances the conservative state

$$\mathbf{U} = [\rho,\ \rho u,\ \rho v,\ \rho E]^T \tag{1}$$

according to the two-dimensional compressible Navier–Stokes equations in conservative form,

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}^i}{\partial x} + \frac{\partial \mathbf{G}^i}{\partial y} = \frac{\partial \mathbf{F}^v}{\partial x} + \frac{\partial \mathbf{G}^v}{\partial y}. \tag{2}$$

The inviscid fluxes are

$$\mathbf{F}^i = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ u(\rho E + p) \end{bmatrix}, \qquad \mathbf{G}^i = \begin{bmatrix} \rho v \\ \rho uv \\ \rho v^2 + p \\ v(\rho E + p) \end{bmatrix}. \tag{3}$$

2.1 Perfect-gas closure

The gas is calorically perfect,

$$p = (\gamma - 1)\rho e, \qquad E = e + \tfrac{1}{2}\left(u^2 + v^2\right), \qquad a = \sqrt{\gamma p/\rho}, \tag{4}$$

with $T = p/(\rho R)$, total enthalpy $H = E + p/\rho$, and $c_p = \gamma R/(\gamma - 1)$. The case files set $\gamma = 1.4$, $R = 1$ and $Pr = 0.72$. Because these are case inputs rather than hard-coded constants, the temperature scale is fixed by the case: the subsonic airfoil cases have $p_\infty = 31.746$ with $\rho_\infty = 1$, hence $T_\infty = 31.746$.

2.2 Nondimensionalization and reference quantities

All cases are posed directly in nondimensional variables through the freestream: $\rho_\infty = 1$, $|\mathbf{u}_\infty| = 1$, and a pressure chosen so that the stated Mach number is reproduced, $p_\infty = 1/(\gamma M_\infty^2)$. The reference length is the chord (airfoil) or the diameter (cylinder), both unity. The solver does not assume this normalization: it reads ρ_∞, $|\mathbf{u}_\infty|$ and p_∞ from the case file, forms $a_\infty = \sqrt{\gamma p_\infty/\rho_\infty}$, and *verifies* that $|\mathbf{u}_\infty|/a_\infty$ matches the stated Mach number to a relative tolerance of 10^{-6}, aborting with a diagnostic if the case file is internally inconsistent. The freestream direction follows the angle of attack, $\hat{\mathbf{d}} = (\cos\alpha, \sin\alpha)$; all benchmark cases use $\alpha = 0$.

The dynamic pressure is $q_\infty = \frac{1}{2}\rho_\infty|\mathbf{u}_\infty|^2$ and force coefficients follow the case reference area A_{ref} and length L_{ref}:

$$C_D = \frac{\mathbf{F}\cdot\hat{\mathbf{d}}}{q_\infty A_{\mathrm{ref}}}, \qquad C_L = \frac{\mathbf{F}\cdot\hat{\mathbf{l}}}{q_\infty A_{\mathrm{ref}}}, \qquad C_{m,z} = \frac{M_z}{q_\infty A_{\mathrm{ref}} L_{\mathrm{ref}}}, \tag{5}$$

with $\hat{\mathbf{l}} = (-\sin\alpha, \cos\alpha)$ the lift direction, so a nonzero angle of attack is handled correctly. The wall distributions reported in `surface.csv` are $C_p = (p - p_\infty)/q_\infty$ and $C_f = \tau_t/q_\infty$, where τ_t is the *tangential* component of the viscous traction (section 5.5).

2.3 Viscous closure

For laminar cases the Newtonian stress tensor is

$$\tau_{xx} = 2\mu\frac{\partial u}{\partial x} - \frac{2}{3}\mu\,\nabla\cdot\mathbf{u}, \quad \tau_{yy} = 2\mu\frac{\partial v}{\partial y} - \frac{2}{3}\mu\,\nabla\cdot\mathbf{u}, \quad \tau_{xy} = \mu\left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}\right), \tag{6}$$

with $\nabla\cdot\mathbf{u} = \partial_x u + \partial_y v$, and the heat flux is Fourier, $\mathbf{q} = -k\nabla T$ with $k = \mu c_p/Pr$. The viscous normal flux assembled at a face with unit normal $\mathbf{n}$ is

$$\mathbf{F}^v\cdot\mathbf{n} = \begin{bmatrix} 0 \\ \tau_{xx}n_x + \tau_{xy}n_y \\ \tau_{xy}n_x + \tau_{yy}n_y \\ u\,(\tau_{xx}n_x + \tau_{xy}n_y) + v\,(\tau_{xy}n_x + \tau_{yy}n_y) + k\,\nabla T\cdot\mathbf{n} \end{bmatrix}. \tag{7}$$

All benchmark cases specify a constant viscosity, obtained directly from the Reynolds number,

$$\mu = \frac{\rho_\infty|\mathbf{u}_\infty|L_{\mathrm{ref}}}{Re}, \tag{8}$$

so $Re = 5000$ gives $\mu = 2\times10^{-4}$, and the cylinder cases give $\mu = 5\times10^{-2}$ ($Re = 20$) and $\mu = 5\times10^{-3}$ ($Re = 200$). A Sutherland law referenced to the freestream temperature is implemented and selectable per case, but no benchmark case uses it. In inviscid mode μ and k are identically zero and the viscous face work is skipped entirely rather than multiplied by zero.

3 Meshes, Boundary Tags, and Case Inputs

3.1 CGNS input

Meshes are read through the CGNS mid-level library. The reader walks every base and every zone, accepts `Unstructured` zones only, and reads the coordinate arrays `CoordinateX/CoordinateY`. Both supplied meshes declare `PhysicalDimension` = 3 with a nominally zero `CoordinateZ`, which is read and discarded. Element sections are scanned for the supported two-dimensional element types `TRI_3` and `QUAD_4`, which become cells, and `BAR_2`, which becomes a boundary edge. No structured-grid or single-element-type assumption is made anywhere: the NACA mesh is genuinely mixed, with 10 752 triangles and 10 064 quadrilaterals.

Two CGNS subtleties required specific handling and are worth recording:

- **Boundary conditions.** The `BC_t` nodes store a `PointRange` with `GridLocation=EdgeCenter`. These indices are *element* indices, not node indices, and they coincide with the `ElementRange` of the identically named `BAR_2` section. The reader therefore maps each boundary condition to its edge section by range containment, then tags the resulting mesh edges with the family name. Family names are carried as strings and matched against the `boundary_conditions` dictionary of the case file, so `bc-2/bc-4` and `WALL/FAR` are handled by the same code path. An unmapped or unknown family is a fatal, clearly reported error.

- **Multi-zone connectivity.** The cylinder mesh has two zones. Its interfaces are stored as `GridConnectivity_t` of type `Abutting1to1`, *not* as `GridConnectivity1to1_t`: a `cg_n1to1` query returns zero. The reader therefore uses `cg_nconns/cg_conn_info/cg_conn_read` and interprets the `PointList/PointListDonor` pairs as one-based zone-local *vertex* indices (the SIDS default location). Paired nodes are merged into single global nodes; the supplied interface point lists are bit-identical, so the maximum merge distance is exactly zero. After merging, the cylinder mesh has 9 995 nodes and 10 185 cells.

3.2 Face extraction and geometry

Cells are converted to an internal polygon representation and a face list is built by hashing each edge's sorted node pair. An edge seen by two cells becomes an interior face with a left and a right cell; an edge seen once becomes a boundary face carrying its family tag. This yields the cell–face adjacency graph used for reconstruction, for the gradient stencils and, in its dual form, for partitioning.

Cell area and centroid come from the shoelace formula over the vertex loop,

$$A = \frac{1}{2}\sum_i \left(x_i y_{i+1} - x_{i+1} y_i\right), \qquad \mathbf{x}_c = \frac{1}{6A}\sum_i (\mathbf{x}_i + \mathbf{x}_{i+1})\left(x_i y_{i+1} - x_{i+1} y_i\right), \tag{9}$$

which is exact for both triangles and quadrilaterals and needs no subdivision. Every cell in both meshes is stored counter-clockwise, so all signed areas are positive; a negative area is treated as a fatal mesh error. In two dimensions a face is a straight segment from $\mathbf{x}_a$ to $\mathbf{x}_b$, so its length Δs (the two-dimensional analogue of face area) and unit normal are

$$\Delta s = \|\mathbf{x}_b - \mathbf{x}_a\|, \qquad \mathbf{n} = \frac{1}{\Delta s}\left(y_b - y_a,\ -(x_b - x_a)\right), \tag{10}$$

with the face centroid at the segment midpoint. Each face normal is then oriented outward from its left cell by testing the sign of $\mathbf{n} \cdot (\mathbf{x}_f - \mathbf{x}_{c,\text{left}})$ and flipping when required. On a boundary face the left cell is by construction the adjacent fluid cell, so **the boundary normal points out of the fluid and into the body**. That convention matters for the force integration in section 5.5 and is the single easiest sign error to make in a solver of this kind.

3.3 Mesh and case summary

Table 2 summarises both meshes. The wall-boundary counts are what the surface output and force integration act on: 404 edges around the airfoil and 100 around the cylinder.

Table 2: Supplied meshes as read and preprocessed by the solver. The cylinder figures are after the two zones have been merged across their `Abutting1to1` interface.

Quantity	NACA0012_H2	CylinderB1
CGNS zones	1	2 (merged)
Nodes	15 682	9 995
Cells	20 816	10 185
triangles	10 752	mixed
quadrilaterals	10 064	mixed
Faces	36 498	20 180
Boundary faces	484	120
Wall-family edges	404 (`bc-4`)	100 (`WALL`)
Farfield edges	80 (`bc-2`)	20 (`FAR`)
Farfield extent	$\approx 80\,c$	$200\,D$
Body extent	$x \in [0, 1.005]$, $t = 0.120$	$r = 0.5$ exactly

A note on the airfoil geometry: the `bc-4` family spans $x \in [0, 1.005]$ with a maximum thickness of 0.120, consistent with a NACA0012 section of unit chord. Its identification as the body, rather than `bc-2` at radius ≈ 80, comes from the case-file mapping and not from geometry.

4 Spatial Discretization

Integrating eq. (2) over a fixed control volume Ω_i and applying the divergence theorem gives the cell-centred semi-discrete form

$$V_i \frac{\mathrm{d}\overline{\mathbf{U}}_i}{\mathrm{d}t} = \mathbf{R}_i = -\sum_{f \in \partial\Omega_i} \Big[\mathbf{H}^i(\mathbf{W}_f^-, \mathbf{W}_f^+, \mathbf{n}_f) - \mathbf{H}^v(\mathbf{W}_f, \nabla\mathbf{W}_f, \mathbf{n}_f) \Big] \Delta s_f, \tag{11}$$

where $\overline{\mathbf{U}}_i$ is the cell average, V_i the cell area, $\mathbf{H}^i$ the numerical inviscid normal flux, $\mathbf{H}^v$ the viscous normal flux of eq. (7), and $\mathbf{n}_f$ the outward unit normal of face f.

The implementation stores each face once and visits it once. Because the stored normal is outward from the left cell, the accumulated flux is subtracted from the left cell and added to the right cell:

$$\mathbf{R}_{\mathrm{left}(f)} \mathrel{-}= \mathbf{H}_f\,\Delta s_f, \qquad \mathbf{R}_{\mathrm{right}(f)} \mathrel{+}= \mathbf{H}_f\,\Delta s_f. \tag{12}$$

A single face therefore contributes equal and opposite amounts to its two neighbours, which makes the discretization conservative by construction: interior contributions telescope exactly and only boundary fluxes can change the global integral. This is verified to 9.2×10^{-17} in section 8.

On a boundary face there is no right cell. The left state is reconstructed to the face as usual and an appropriate exterior state is synthesised by the boundary condition (section 5); the same numerical flux function is then applied. This keeps boundary and interior faces on one code path, so no boundary case can drift out of sync with the interior scheme.

A face is processed if either of its cells is owned by the current rank. Faces on a partition boundary are thus evaluated redundantly on both ranks, which is cheaper than communicating fluxes and guarantees that the two ranks compute bit-identical values from identical inputs.

4.1 Second-Order Reconstruction

Second-order accuracy comes from a piecewise-linear reconstruction of the *primitive* variables $\mathbf{W} = (\rho, u, v, p)$. Primitives are chosen so that the positivity of density and pressure can be enforced directly on the reconstructed face state, which is not possible when limiting conservative variables.

For each cell the gradient is an inverse-distance-weighted least-squares fit over the face neighbours. The gradient $\mathbf{g}_i$ of a variable ϕ minimises

$$J(\mathbf{g}_i) = \sum_{j\in\mathcal{N}(i)} w_j^2 \left[\mathbf{g}_i\cdot(\mathbf{x}_j - \mathbf{x}_i) - (\phi_j - \phi_i)\right]^2, \qquad w_j = \frac{1}{\|\mathbf{x}_j - \mathbf{x}_i\|}, \tag{13}$$

whose normal equations are the 2×2 system

$$\underbrace{\begin{bmatrix} \sum_j w_j^2 \Delta x_j^2 & \sum_j w_j^2 \Delta x_j \Delta y_j \\ \sum_j w_j^2 \Delta x_j \Delta y_j & \sum_j w_j^2 \Delta y_j^2 \end{bmatrix}}_{\mathbf{A}_i} \mathbf{g}_i = \sum_j w_j^2 (\phi_j - \phi_i) \begin{bmatrix} \Delta x_j \\ \Delta y_j \end{bmatrix}. \tag{14}$$

Since $\mathbf{A}_i$ depends only on geometry, it is inverted once during preprocessing and folded into a per-neighbour weight vector $\mathbf{W}_j = w_j^2\, \mathbf{A}_i^{-1} \Delta\mathbf{x}_j$, so that at run time the gradient of every variable is the single sum $\mathbf{g}_i = \sum_j \mathbf{W}_j(\phi_j - \phi_i)$ — one dot product per neighbour per variable, with no linear algebra in the hot loop. A nearly rank-deficient stencil (possible when a cell's neighbours are almost collinear) is detected by comparing $\det \mathbf{A}_i$ against $10^{-12}\,\mathrm{tr}(\mathbf{A}_i)^2$ and regularised by a small diagonal shift, rather than being allowed to produce an unbounded gradient.

Boundary faces contribute a stencil entry located at the face centroid and carrying the *physical* boundary state (section 5), not the mirrored ghost state. This distinction is essential: using a mirrored state here would double every wall-normal gradient and corrupt the skin friction in the first cell off the wall.

The face states are then

$$\mathbf{W}_f^{\pm} = \mathbf{W}_i + \Phi_i \odot \left[\mathbf{g}_i\cdot(\mathbf{x}_f - \mathbf{x}_i)\right], \tag{15}$$

with Φ_i the limiter of section 4.2 and $\odot$ a component-wise product. Second-order reconstruction is **active in every production run reported here**; the only first-order fallback is the positivity mechanism below, whose activation count is recorded per run and reported in `metadata.json`.

4.2 Limiter and Positivity Control

Both a Barth–Jespersen and a Venkatakrishnan limiter are implemented; the production runs use Venkatakrishnan. For each cell and variable the neighbour envelope $\phi_i^{\min}, \phi_i^{\max}$ is gathered over the face neighbours, including boundary-face states so that a wall-adjacent cell is not falsely flagged as an extremum. With Δ_f the unlimited increment towards face f from eq. (15) and $y = (\phi_i^{\max} - \phi_i)/\Delta_f$ for $\Delta_f > 0$ (or $y = (\phi_i^{\min} - \phi_i)/\Delta_f$ for $\Delta_f < 0$), the two limiters are

$$\Phi^{\mathrm{BJ}} = \min(1,\, y), \qquad \Phi^{\mathrm{V}} = \frac{y^2 + 2y + \epsilon^2/\Delta_f^2}{y^2 + y + 2 + \epsilon^2/\Delta_f^2}, \tag{16}$$

and the cell limiter is the minimum over its faces, clipped to $[0, 1]$. The Venkatakrishnan threshold is $\epsilon^2 = (Kh_i)^3$ with $h_i = \sqrt{V_i}$ the cell length scale and K a case-settable constant. This threshold is what switches the limiter off in smooth regions: without it, limiter chattering on a stretched mesh stalls the steady residual, which is precisely the mechanism discussed in section 8.5.

Positivity is protected in two places. At the reconstruction stage, if the limited face state still has $\rho \leq 0$ or $p \leq 0$, the whole increment is halved repeatedly (up to eight times) and finally dropped to the first-order cell average, which is always admissible; each such event is counted. After each implicit update a final safety net floors density and pressure at 10^{-8} of their freestream values and replaces any non-finite state by the freestream, again counting every occurrence. Because these counts are written to the run metadata, a run cannot quietly rely on clipping to look healthy.

4.3 Inviscid and Viscous Fluxes

Three inviscid numerical fluxes are implemented — Roe with an entropy fix, HLLC, and Rusanov/local Lax–Friedrichs — selectable per run. Production runs use **Roe**. With Roe averages $\hat{\rho} = \sqrt{\rho_L \rho_R}$ and $\hat{\phi} = (\sqrt{\rho_L}\phi_L + \sqrt{\rho_R}\phi_R)/(\sqrt{\rho_L} + \sqrt{\rho_R})$ for $\phi \in \{u, v, H\}$, and $\hat{a}^2 = (\gamma - 1)(\hat{H} - \frac{1}{2}|\hat{\mathbf{u}}|^2)$, the flux is

$$\mathbf{H}^{\mathrm{Roe}} = \tfrac{1}{2}(\mathbf{F}_L + \mathbf{F}_R) - \tfrac{1}{2}\sum_k \tilde{\lambda}_k \alpha_k \mathbf{r}_k, \tag{17}$$

with wave speeds $\lambda_{1,2,3} = \hat{u}_n - \hat{a},\ \hat{u}_n,\ \hat{u}_n + \hat{a}$, acoustic amplitudes $\alpha_{1,3} = (\Delta p \mp \hat{\rho}\hat{a}\Delta u_n)/(2\hat{a}^2)$, entropy amplitude $\alpha_2 = \Delta\rho - \Delta p/\hat{a}^2$, and a separately assembled shear contribution proportional to $\hat{\rho}\,\Delta u_t$.

Entropy fix. The eigenvalues are smoothed by the Harten function

$$\tilde{\lambda}(\lambda, \delta) = \begin{cases} |\lambda|, & |\lambda| \geq \delta, \\ \dfrac{\lambda^2 + \delta^2}{2\delta}, & |\lambda| < \delta, \end{cases} \tag{18}$$

applied to *all three* waves. For the acoustic waves δ is the Harten–Hyman spread, e.g. $\delta_1 = \max(0, \lambda_1 - (u_{n,L} - a_L), (u_{n,R} - a_R) - \lambda_1)$, which removes the non-physical expansion shock admitted by plain Roe at a sonic point. In addition, every wave receives a floor

$$\delta \geq \delta_{\min} = 0.05\ \max(|u_{n,L}| + a_L,\ |u_{n,R}| + a_R)\,. \tag{19}$$

The floor addresses a second, distinct failure that proved important on these meshes. The entropy and shear waves share the eigenvalue u_n, which vanishes wherever the flow is parallel to a face — at a stagnation point, and along the symmetry line of a symmetric body. There the density and tangential-velocity jumps are left completely undamped, which is the classical Roe checkerboard/carbuncle mode. It does not diverge; it quietly stalls the steady residual and breaks the symmetry that a zero-incidence case should preserve. Applying the floor through the same smooth Harten function keeps the flux consistent, because identical left and right states make the jumps themselves vanish. The coefficient 0.05 is small enough to leave boundary layers and contact discontinuities sharp.

Viscous fluxes. Face gradients are built from the same least-squares cell gradients. On an interior face the two cell gradients are averaged and then corrected along the cell-to-cell direction $\mathbf{e} = \Delta\mathbf{x}/\|\Delta\mathbf{x}\|$ so the face gradient reproduces the actual cell-value difference exactly,

$$\nabla\phi_f = \overline{\nabla\phi} + \left(\frac{\phi_R - \phi_L}{\|\Delta\mathbf{x}\|} - \overline{\nabla\phi}\cdot\mathbf{e}\right)\mathbf{e}, \tag{20}$$

which suppresses the odd-even decoupling that a plain average allows. The temperature gradient is derived from the density and pressure gradients through $T = p/(\rho R)$, $\nabla T = R^{-1}\left(\nabla p/\rho - p\,\nabla\rho/\rho^2\right)$.

On a wall face the viscous flux is evaluated with the *physical* wall state and the wall-normal derivative is corrected using the imposed wall value: with $d_n = (\mathbf{x}_f - \mathbf{x}_i)\cdot\mathbf{n}$,

$$\nabla\phi \leftarrow \nabla\phi + \left(\frac{\phi_{\text{wall}} - \phi_i}{d_n} - \nabla\phi\cdot\mathbf{n}\right)\mathbf{n}. \tag{21}$$

This is what makes the skin friction reflect the actual near-wall profile on a stretched boundary-layer mesh instead of the much smaller cell-average gradient. For an adiabatic wall the normal component of ∇T is then projected out exactly, enforcing $\partial T/\partial n = 0$.

Positivity fallback. If the Roe average is inadmissible ($\hat{a}^2 \leq 0$, possible for a very strong expansion), the flux falls back to HLLC for that face, which is positivity preserving. This is reported in `metadata.json` as part of the flux description rather than hidden. In practice this path is never taken in any production run reported here: every case records zero positivity-fallback events. That has a consequence for what the production results can and cannot verify, discussed in section 8.6.

Named dissipation and limiter parameters. For completeness, the scheme carries four tuned constants, all disclosed here rather than buried in the source: the linear-wave dissipation floor coefficient 0.05 in eq. (19); the Venkatakrishnan threshold constant $K = 5.0$ in eq. (16); and the viscous spectral-radius factors 4.0 in the local time step eq. (29) and 2.0 in the implicit off-diagonal eq. (32).

4.3.1 Added dissipation on the linear waves: disclosure

The floor eq. (19) deserves to be stated plainly, because only part of it is textbook. Smoothing the *acoustic* eigenvalues by the Harten–Hyman spread is the standard entropy fix and needs no defence. Applying a floor to the *entropy and shear* eigenvalues is different: it is added artificial dissipation, a carbuncle cure, and it is not part of the classical Roe scheme.

It is present because without it these cases do not converge correctly. The entropy and shear waves share the eigenvalue u_n, which vanishes identically on any face where the flow is parallel to the face — including the stagnation streamline of a symmetric body at zero incidence, which is exactly the configuration of all six airfoil cases. There, density and tangential-velocity jumps are advected with zero dissipation, producing a checkerboard mode that does not diverge but stalls the residual and destroys the symmetry the solution should have.

The cost is quantifiable. At a stagnation face where the exact numerical flux is $[0, p, 0, 0]$ in normalised form, the floored scheme returns $[-2.96\times10^{-4}, 1.0, 2.13\times10^{-4}, 7.42\times10^{-5}]$: a mass-flux error of about 3×10^{-4} relative to the pressure term. The scheme remains consistent, since identical left and right states make all jumps vanish and the exact physical flux is recovered, but at a stagnation point the added dissipation is genuinely there.

The honest concern is that at $Re = 5000$ this numerical dissipation competes with the physical viscosity, so a fraction of the computed skin friction could in principle be an artefact of the floor rather than of μ. That concern was tested rather than argued away, by sweeping the coefficient in controlled experiments. Section 9.1 reports the outcome in full, and it does not support the design choice: the floor changes C_D by 0.021 % on the cylinder, and disabling it entirely produces no checkerboard mode even on the zero-incidence inviscid airfoil, where the entropy and shear eigenvalues vanish on the stagnation streamline and no physical viscosity is present to damp a carbuncle. The floor is therefore retained as a safeguard for conditions not exercised here, and no result in this report depends on it.

5 Boundary Conditions

Three boundary conditions are implemented, selected per family by the case file: `farfield`, `slip_wall`, and `no_slip_adiabatic_wall`.

5.1 Ghost state versus face state

A design point worth stating explicitly, because it is a common source of silent error, is that each boundary condition supplies *two* different exterior states and the solver uses them in different places:

- The **ghost state** $\mathbf{U}^{\text{ghost}}$ is a mirrored or characteristic state passed to the Riemann solver as the right state. Its purpose is that the resulting numerical flux *enforces* the boundary condition.
- The **face state** $\mathbf{U}^{\text{face}}$ is the physical state that actually exists on the boundary. It is used for the least-squares gradient stencil, for the limiter envelope, for the viscous wall flux, and for the surface output written to `surface.csv`.

Conflating the two is what produces the classic artefacts: doubled wall-normal gradients, wall rows in the surface output that are really adjacent cell-centre values, and a no-slip wall that reports a nonzero velocity. The distinction also means `surface.csv` carries genuine boundary values, which is declared in the metadata as `wall_boundary_output_semantics = boundary_value`. Cell-centre quantities are written separately to `surface_cell_center.csv` so the two can never be confused.

5.2 Inviscid slip wall

The ghost state reflects the normal momentum component while preserving density and total energy,

$$\mathbf{u}^{\text{ghost}} = \mathbf{u} - 2(\mathbf{u}\cdot\mathbf{n})\mathbf{n}, \qquad \rho^{\text{ghost}} = \rho, \qquad (\rho E)^{\text{ghost}} = \rho E, \tag{22}$$

which is consistent because reflection leaves $|\mathbf{u}|$ unchanged. Feeding eq. (22) to the Roe flux gives, for identical left and right thermodynamic states, exactly

$$\mathbf{H}^{\text{wall}} = [0,\ p\,n_x,\ p\,n_y,\ 0]^T\,, \tag{23}$$

i.e. zero mass flux, zero energy flux, and a pure pressure force. This identity is checked numerically as part of the verification suite (section 8).

The corresponding face state removes the normal velocity entirely, $\mathbf{u}^{\text{face}} = \mathbf{u} - (\mathbf{u}\mathbf{n})\mathbf{n}$, keeps ρ and p, and corrects the total energy for the reduced kinetic energy. A slip wall therefore reports near-zero *normal* velocity with a retained tangential velocity, exactly as the output contract requires.

5.3 Viscous no-slip adiabatic wall

The ghost state reverses the entire velocity vector, $\mathbf{u}^{\text{ghost}} = -\mathbf{u}$, with ρ and ρE unchanged, so that the arithmetic face average of the interior and ghost velocities is exactly zero. The face state imposes the wall condition directly: zero velocity, interior pressure ($\partial p/\partial n = 0$), and interior temperature (adiabatic, $\partial T/\partial n = 0$), with the density following from the equation of state so that the wall state is thermodynamically consistent rather than an ad-hoc copy,

$$\mathbf{u}^{\text{face}} = \mathbf{0}, \quad p^{\text{face}} = p_i, \quad T^{\text{face}} = T_i, \quad \rho^{\text{face}} = \frac{p^{\text{face}}}{R\,T^{\text{face}}}, \quad (\rho E)^{\text{face}} = \frac{p^{\text{face}}}{\gamma - 1}. \tag{24}$$

Because the surface output uses this state, the reported wall velocity and Mach number are identically zero to machine precision. The adiabatic condition is additionally enforced on the flux by projecting the normal component out of ∇T, as described after eq. (21).

5.4 Farfield

The farfield is characteristic, based on locally one-dimensional Riemann invariants along the face normal. With interior values $(\rho_i, p_i, u_{n,i}, a_i)$ and freestream values $(\rho_\infty, p_\infty, u_{n,\infty}, a_\infty)$, the normal Mach number $M_n = u_{n,i}/a_i$ selects the branch:

- $M_n \geq 1$ (supersonic outflow): the boundary state is the interior state; nothing is imposed.
- $M_n \leq -1$ (supersonic inflow): the boundary state is the freestream.
- $|M_n| < 1$ (subsonic): the outgoing and incoming invariants

$$R^+ = u_{n,i} + \frac{2a_i}{\gamma - 1}, \qquad R^- = u_{n,\infty} - \frac{2a_\infty}{\gamma - 1} \tag{25}$$

  are combined into $u_{n,b} = \frac{1}{2}(R^+ + R^-)$ and $a_b = \frac{1}{4}(\gamma - 1)(R^+ - R^-)$. Entropy $s = p/\rho^\gamma$ and the tangential velocity are taken from the upwind side (interior if $u_{n,b} > 0$, freestream otherwise), giving $\rho_b = \left(a_b^2/(\gamma s)\right)^{1/(\gamma-1)}$ and $p_b = s\rho_b^\gamma$.

This treatment is what allows the supersonic cases to be posed on the same mesh as the subsonic ones without special handling: at $M_\infty = 2$ the inflow arc is fully determined by the freestream and the outflow arc is fully determined by the interior, so the bow shock can leave the domain cleanly. Degenerate invariant combinations that would give $a_b \leq 0$, or a non-positive reconstructed density or pressure, fall back to the freestream state rather than producing an unphysical boundary value. The same characteristic state is used as both ghost and face state at the farfield, since no wall condition applies there.

5.5 Force integration and the pressure/friction split

Forces are integrated over wall faces only. Recalling from section 3 that the stored boundary normal $\mathbf{n}$ points out of the fluid and into the body, the outward normal of the *body* surface is $\mathbf{m} = -\mathbf{n}$, and the Cauchy traction exerted by the fluid on the body is

$$\mathbf{t} = \boldsymbol{\sigma}\cdot\mathbf{m} = (-p\mathbf{I} + \boldsymbol{\tau})\cdot(-\mathbf{n}) = p\,\mathbf{n} - \boldsymbol{\tau}\cdot\mathbf{n}. \tag{26}$$

The pressure force per face is therefore $+(p_{\text{wall}} - p_\infty)\mathbf{n}\,\Delta s$ — referenced to p_∞ so that a uniform ambient pressure contributes nothing on a closed body — and the viscous force is $-\boldsymbol{\tau}\cdot\mathbf{n}\,\Delta s$. Getting this sign pair right is essential: with the opposite viscous sign a flat plate with $\partial u/\partial y > 0$ would produce thrust instead of drag.

The wall pressure is reconstructed from the adjacent cell with the same gradient used by the flux, so the integrand is second-order accurate and consistent with the residual. For the friction the *tangential* part of the viscous traction is used,

$$\boldsymbol{\tau}_t = \boldsymbol{\tau}\cdot\mathbf{n} - [(\boldsymbol{\tau}\cdot\mathbf{n})\cdot\mathbf{n}]\,\mathbf{n}, \tag{27}$$

which is what the contract requires: the normal viscous traction is explicitly excluded rather than being mislabelled as skin friction. The wall-normal velocity derivatives entering $\boldsymbol{\tau}$ are corrected with the imposed zero wall velocity exactly as in eq. (21).

Each wall face is integrated by the rank that owns its adjacent cell, so a face on a partition boundary is counted exactly once globally, and the six accumulators (pressure force, viscous force, moment, wall length) are combined in a single `MPI_Allreduce`. The moment about the case reference centre is $M_z = \sum(\mathbf{r}\times\mathbf{f})_z$ with $\mathbf{r} = \mathbf{x}_f - \mathbf{x}_{\text{ref}}$, positive counter-clockwise.

6 Implicit and Transient Time Integration

6.1 Steady pseudo-time march

Steady cases solve $\mathbf{R}(\mathbf{U}) = \mathbf{0}$ by a backward-Euler pseudo-time march. Linearising eq. (11) about the current iterate gives, at each pseudo-time step, the linear system

$$\left(\frac{V_i}{\Delta\tau_i}\mathbf{I} + \frac{\partial\mathbf{R}}{\partial\mathbf{U}}\right)\Delta\mathbf{U} = \mathbf{R}(\mathbf{U}^m), \qquad \mathbf{U}^{m+1} = \mathbf{U}^m + \Delta\mathbf{U}. \tag{28}$$

The pseudo-time step is local to each cell and set from the convective and viscous spectral radii accumulated during the residual evaluation,

$$\Delta\tau_i = \frac{\mathrm{CFL}\cdot V_i}{\displaystyle\sum_{f\in\partial\Omega_i} (|u_n| + a)_f\,\Delta s_f \;+\; C_v \sum_{f\in\partial\Omega_i} \frac{\mu_f}{\rho_f}\frac{\Delta s_f^2}{V_i}}, \tag{29}$$

with $C_v = 4$ for the viscous contribution. Local time stepping lets the tiny wall cells and the huge farfield cells advance at their own stable rates, which is what makes a mesh with a cell-volume range of roughly ten orders of magnitude tractable at all.

The CFL number is ramped geometrically in log space from `cfl_initial` to `cfl_max` over `pseudo_cfl_ramp_steps`,

$$\mathrm{CFL}(m) = \mathrm{CFL}_0\left(\frac{\mathrm{CFL}_{\max}}{\mathrm{CFL}_0}\right)^{m/M_{\text{ramp}}}, \tag{30}$$

capped at $\mathrm{CFL}_{\max}$ thereafter. The supplied case values are used without modification; table 3 lists them.

A residual-based safeguard multiplies the ramped CFL by an adaptive factor. If a step increases the residual by more than 30%, or produces a non-finite residual, the step is rejected, the previous accepted state is restored, and the factor is cut to 0.35 of its value; after a successful step the factor recovers by 6% up to unity. Rejected-step counts are reported. This is a robustness device, not an accuracy device: it changes only the path to the steady state.

6.2 LU-SGS inner solve and the Jacobian approximation

Equation (28) is solved approximately by symmetric Gauss–Seidel sweeps (LU-SGS); a block-Jacobi variant is also available. The Jacobian is never assembled. The diagonal is the scalar

$$D_i = \frac{V_i}{\Delta \tau_i} + \frac{1}{2} \sum_{f \in \partial \Omega_i} (|u_n| + a)_f \, \Delta s_f + \sum_{f \in \partial \Omega_i} \left(\frac{\mu}{\rho} \right)_f \frac{\Delta s_f^2}{V_i}, \tag{31}$$

where the factor $\frac{1}{2}$ is the upwind split of the face dissipation between the two adjacent cells. The off-diagonal action on a neighbour is the scalar-dissipation upwind approximation

$$\mathbf{O}_{ij} \Delta \mathbf{U}_j = \frac{\Delta s_f}{2} \left[\mathbf{A}_j(\mathbf{n}) \Delta \mathbf{U}_j - \lambda_j \Delta \mathbf{U}_j \right], \qquad \lambda_j = |u_{n,j}| + a_j + \frac{2\mu}{\rho_j \, \|\Delta \mathbf{x}\|}, \tag{32}$$

with the last term present only for viscous runs. Crucially, $\mathbf{A}_j(\mathbf{n}) \Delta \mathbf{U}_j$ is the *exact analytic* Jacobian-vector product of the Euler normal flux, evaluated matrix-free: no 4×4 block is ever stored. Writing $\mathrm{d}(\rho u_n) = u_n \, \mathrm{d}\rho + \rho \, \mathrm{d}u_n$ and $\mathrm{d}p = (\gamma - 1) \left[\mathrm{d}(\rho E) - \frac{1}{2} |\mathbf{u}|^2 \mathrm{d}\rho - \rho(u \, \mathrm{d}u + v \, \mathrm{d}v) \right]$, the product is

$$\mathbf{A}(\mathbf{n}) \, \mathrm{d}\mathbf{U} = \begin{bmatrix} \mathrm{d}(\rho u_n) \\ \mathrm{d}(\rho u_n) u + \rho u_n \, \mathrm{d}u + \mathrm{d}p \, n_x \\ \mathrm{d}(\rho u_n) v + \rho u_n \, \mathrm{d}v + \mathrm{d}p \, n_y \\ \mathrm{d}u_n (\rho E + p) + u_n \left(\mathrm{d}(\rho E) + \mathrm{d}p \right) \end{bmatrix}. \tag{33}$$

This product is verified against central finite differences to 1.7×10^{-8} over 20 000 randomised states (section 8).

Each sweep refreshes the ghost increments once, then performs a forward pass over ascending local cell index and a backward pass over descending index. Between refreshes the off-rank coupling is frozen, which makes the scheme exactly block-Jacobi across ranks and pure LU-SGS within a rank — so the iteration count can depend mildly on the partition even though the converged answer does not.

The inner iteration is monitored by the true linear residual of the system it is solving,

$$r_{\mathrm{lin}} = \frac{\|\mathbf{R} - (D \, \Delta \mathbf{U} + \mathbf{O} \, \Delta \mathbf{U})\|_2}{\|\mathbf{R}\|_2}, \tag{34}$$

evaluated with the same matrix-free operator the sweeps use and volume-normalised like the non-linear norms. Sweeps are added in small blocks, within the case-file bounds, until r_{lin} meets the case target or fails to improve by more than 2% in three consecutive blocks. The increment is deliberately *not* reset between blocks — an early version did reset it, which silently discarded all previous work and capped the achievable inner convergence.

One negative result is worth recording, because it is a tempting shortcut. Inflating the diagonal by a factor $\beta > 1$ makes the sweeps look dramatically better: at $\beta = 2$ the reported ratio reached 2.2×10^{-16}. That number is meaningless, because it measures a different (easier) system than the one being solved; the true residual ratio of eq. (34) was 0.82. The diagonal is therefore left at its consistent value eq. (31), and the slow-but-honest convergence of SGS at high CFL is accepted instead.

6.3 Convergence criteria

All reported residual norms are volume-weighted root-mean-square rates of change of the cell average, globally MPI-reduced:

$$\|\mathbf{R}\|_2 = \left(\frac{\sum_i \sum_k (R_{i,k}/V_i)^2 V_i}{\sum_i V_i} \right)^{1/2}, \qquad \|\mathbf{R}\|_\infty = \max_{i,k} |R_{i,k}/V_i| . \tag{35}$$

Dividing by the cell volume converts the integral residual into a rate, which makes the norm independent of local cell size and therefore comparable across meshes and rank counts. Section 8.5 describes the termination test built on these norms, and the defect that had to be fixed in it.

6.4 Transient dual-time BDF2 for the vortex street

The $Re = 200$ cylinder is integrated in physical time by a genuine two-loop dual-time scheme. The outer loop steps physical time with second-order backward differences; the inner loop drives the resulting nonlinear system to convergence at each physical step.

With $a_0 = \frac{3}{2},\ a_1 = -2,\ a_2 = \frac{1}{2}$ the BDF2 residual is

$$\mathbf{R}^*(\mathbf{U}^{n+1}) = \mathbf{R}(\mathbf{U}^{n+1}) - \frac{V_i}{\Delta t} \left(a_0 \mathbf{U}^{n+1} + a_1 \mathbf{U}^n + a_2 \mathbf{U}^{n-1} \right), \tag{36}$$

and the inner iteration solves $\mathbf{R}^* = \mathbf{0}$ with the same LU-SGS machinery, using the diagonal

$$D_i^* = \frac{V_i}{\Delta \tau_i} + \frac{a_0 V_i}{\Delta t} + \frac{1}{2} \sum_f (|u_n| + a)_f \, \Delta s_f + \ldots, \tag{37}$$

i.e. carrying *both* the pseudo-time and the physical-time term, so each inner iteration is a Newton-like step on the time-accurate system rather than a pseudo-time march with a source term. The first physical step uses the self-starting BDF1 coefficients $a_0 = 1, a_1 = -1, a_2 = 0$.

Two properties required by the benchmark hold explicitly in the implementation:

1. $\mathbf{U}^n$ and $\mathbf{U}^{n-1}$ are stored in separate arrays and are **frozen throughout all inner iterations** for $\mathbf{U}^{n+1}$. Only the $\mathbf{U}^{n+1}$ iterate changes inside the inner loop.

2. The history is shifted ($\mathbf{U}^{n-1} \leftarrow \mathbf{U}^n$, $\mathbf{U}^n \leftarrow \mathbf{U}^{n+1}$) only **after** the inner solve for that physical step has been accepted.

The inner loop terminates when the total transient residual $\|\mathbf{R}^*\|_2$ has fallen by the case target relative to its value at the first inner iteration of that physical step, subject to the minimum iteration count. The convergence measure is the *total* residual including the physical-time term, as the benchmark specifies, not the spatial residual alone. Per-step inner counts, target misses and the final ratio are accumulated and reported in `metadata.json`.

The run is declared `statistically_periodic` only if it reaches the requested horizon *and* the inner solve converged on at least 95% of physical steps. A run that reaches the final time while frequently missing the inner target is an under-converged dual-time solve and is reported as `failed` rather than presented as a usable transient.

For the transient case the supplied production settings are used exactly: $\Delta t = 0.01$, $t_f = 300.0$ (30 000 physical steps), inner iteration bounds 5–1000, inner target 10^{-3} on the total transient residual, and a fixed pseudo-time CFL of 1.0.

Table 3: Production run controls. Every value is taken from the supplied case file and used unmodified; no case uses a loosened setting.

Case group	Max steps	CFL_0	CFL_{max}	Ramp steps	Inner
NACA inviscid $M_\infty = 0.15$	20 000	1.0	100	2000	3–50
NACA inviscid $M_\infty = 0.80$	30 000	1.0	100	3000	3–50
NACA inviscid $M_\infty = 2.0$	40 000	0.2	50	5000	3–80
NACA laminar $M_\infty = 0.15$	40 000	1.0	100	5000	3–80
NACA laminar $M_\infty = 0.80$	40 000	0.5	80	5000	3–80
NACA laminar $M_\infty = 2.0$	50 000	0.2	50	7000	3–100
Cylinder $Re = 20$	30 000	1.0	100	3000	3–80
Cylinder $Re = 200$	30 000	1.0	1.0	0	5–1000

7 MPI Parallelization and METIS Partitioning

7.1 Cell dual graph and METIS

The mesh is read and partitioned once, during setup. The partitioning graph is the *cell dual graph*: each cell is a vertex, and every interior face contributes one undirected edge between its two cells. Boundary faces contribute no edge. The graph is built in compressed adjacency form (`xadj`, `adjncy`) by counting degrees, prefix-summing, then filling.

Partitioning calls `METIS_PartGraphKway` with $n_{\text{con}} = 1$, zero-based numbering, a fixed random seed of 12 345 so that a clean checkout reproduces the partition exactly, and `METIS_OPTION_CONTIG` requested to prefer contiguous parts. If the contiguity constraint causes METIS to fail, the call is retried without it and the fallback is logged; a genuine METIS failure is a fatal error. The reported objective value is the edge cut recorded in `metadata.json`. This is real graph partitioning, not a geometric split: the metadata value `metis_kway` corresponds directly to this call.

7.2 Rank-local mesh and ghost construction

After partitioning, each rank builds a mesh containing only what it needs:

- its **owned** cells, numbered $[0, N_{\text{own}})$;
- a layer of **ghost** cells, numbered $[N_{\text{own}}, N_{\text{own}} + N_{\text{ghost}})$, being exactly those off-rank cells that share a face with an owned cell;
- the faces incident on any owned cell, with left/right indices remapped to local numbering;
- the geometry, gradient stencils and boundary tags for those entities.

No rank ever stores the global cell list, the global face list, or the global conservative state during iterations. Correspondingly, `metadata.json` reports `full_state_replication_during_iterations` and `full_mesh_replication_during_iterations` as `false`.

Ghost ownership is resolved once at setup by a single collective exchange of the requested global cell identifiers, after which each rank knows precisely which of its owned cells to send to which neighbour, in an order shared by both sides. That shared ordering is what allows the iteration-time exchange to be a plain contiguous buffer copy with no index metadata on the wire. This collective appears only in preprocessing; it is not part of the iteration loop.

7.3 Halo exchange

Iteration-time communication is neighbour-scoped point-to-point. For each exchange the receives are posted first with `MPI_Irecv`, the send buffers are packed and sent with `MPI_Isend`, a single `MPI_Waitall` completes all of them, and the received data is unpacked into the ghost slots. The metadata field `halo_exchange` is `neighbor_isend_irecv`. There is no `MPI_Allgather` of state at any point in the iteration.

Each residual evaluation performs a fixed, small number of exchanges:

1. the conservative state (once per residual evaluation, by the driver);
2. the primitive gradients, $4 \times 2 = 8$ components, needed for reconstruction from the far side of a partition face and for the viscous face gradient;
3. the limiter factors, 4 components, so a partition-boundary face reconstructs exactly as it would in serial.

The last two are the reason a partitioned second-order run reproduces the serial answer: the far-side cell has not only its state but also its gradient and its limiter value, so eq. (15) evaluates identically on both sides of a cut. Each LU-SGS sweep additionally exchanges the current increment $\Delta\mathbf{U}$.

Global reductions are used only where they are physically required: one `MPI_Allreduce` for the residual norms (component sums, volume, and a max for the L_∞ norm), one for the six force accumulators, and one for the linear-residual ratio of eq. (34). Output is written by rank 0 from reduced data, so file contents do not depend on rank count except through floating-point summation order.

7.4 Partition diagnostics

Table 4 reports the per-rank decomposition actually used during the iterations of a representative production run (NACA0012 $M_\infty = 0.15$ inviscid), taken from the submitted `partition_diagnostics.csv`. The load-balance ratio, defined as $\max_r N_{\text{own},r}/\overline{N_{\text{own}}}$, is 1.002.

Table 4: Per-rank partition diagnostics for NACA0012 $M_\infty = 0.15$ inviscid, from the submitted `partition_diagnostics.csv`. Send and receive counts are the number of cell records moved per halo exchange.

Rank	Owned	Ghost	Boundary faces	Neighbours	Send	Recv
0	5214	114	119	3	115	114
1	5189	104	129	2	103	104
2	5207	111	125	2	110	111
3	5206	143	111	3	144	143

8 Output, Reproducibility, and Sanity Checks

8.1 Build and run

The solver depends on MPI, CGNS (with HDF5 and zlib), and METIS, plus a header-only JSON reader. It is built with CMake:

```
cmake -S . -B build -DCMAKE_BUILD_TYPE=Release \
      -DCFD_EXTERNALS_ROOT=/opt/external/cfd_externals/install \
      -DCFD_EXTERNALS_HEADER_ROOT=/opt/external
cmake --build build -j 32
```

and every case is run through the single required CLI form:

```
mpirun -np <ranks> ./build/cns2d solve --case <case.json> \
       --output <results-dir> [--restart <file>] [--report-level brief|full]
```

No case file or source edit is needed between cases. Debug-only overrides (`--max-steps`, `--flux`, `--limiter`, `--spatial-order`, and similar) exist for diagnostics; none of them is used in the production runs reported here, and the exact command line for each run is recorded in its `run_status.json`. Malformed input, a missing mesh, an unmapped boundary family, or an inconsistent freestream cause a non-zero exit with a specific message rather than a silent default.

Every flag advertised in the required CLI is implemented. `--report-level` selects how much of the mesh-verification report is printed: `full` lists every measured quantity, while `brief` prints a one-line summary. The verification checks themselves always run, so the level cannot influence a computed result — confirmed by running the same case at both levels and diffing the outputs, which are byte-identical in `residuals.csv` and `forces.csv`. Exit codes are distinct and reduced across ranks with `MPI_MAX`: 0 on success, 1 for an `MPI_Init` failure, 2 for bad input, and 3 for a numerically failed run.

Build and run instructions, dependency versions, per-case outputs, the test suite, the Python tooling and a ten-step reproduce-from-clean-checkout sequence are documented in `README.md` at the root of the solver tree.

8.2 Generated files

Each case directory contains `metadata.json`, `run_status.json`, `residuals.csv`, `forces.csv`, `surface.csv`, `partition_diagnostics.csv` and `.json`, `field_final.vtu`, `restart_final.bin`, and `stdout.log`; viscous cases add `surface_cell_center.csv` carrying the adjacent cell-centre values that `surface.csv` deliberately does not mix in. The report artifacts (`run_manifest.csv`, `figure_manifest.csv`, `sanity_checks.json`) and every figure are regenerated from those files alone by

```
.venv/bin/python tools/make_figures.py --results-root results \
    --out-dir report/figures --manifest report/figure_manifest.csv
```

so no figure can show anything other than submitted data. The numeric values quoted in this report are likewise harvested mechanically from the same files by `report/harvest_numbers.py`, which emits the LaTeX macros and table rows used above; the prose contains no hand-typed result numbers.

8.3 Geometric and discrete verification

The solver verifies its own mesh metrics at startup and reports the results in `stdout.log` before spending time on a solve. Table 5 collects these measurements together with two offline probes.

Three of these deserve comment.

Table 5: Verification measurements. The first four are geometric identities that must hold to machine precision; the remainder test the discretization itself.

Check	Identity tested	Measured	Expected
Face closure	$\sum_f \mathbf{n}_f \Delta s_f = \mathbf{0}$ per cell	1.0×10^{-16}	$O(\varepsilon)$
Volume closure	$\frac{1}{2}\sum_f (\mathbf{x}_f \cdot \mathbf{n}_f)\Delta s_f = V_i$	1.1×10^{-10}	$O(\varepsilon)$
Domain area	$\sum_i V_i$ vs boundary line integral	1.4×10^{-14}	$O(\varepsilon)$
LSQ exactness	gradient of a linear field reproduced exactly	3.1×10^{-13}	$O(\varepsilon)$
Conservation†	$\sum_i \mathbf{R}_i$ vs net boundary flux	9.2×10^{-17}	$O(\varepsilon)$
Jacobian product	eq. (33) vs central finite differences	1.7×10^{-8}	$O(\sqrt{\varepsilon})$
Slip-wall flux	$[0, pn_x, pn_y, 0]$ exactly	exact	exact

†Measured in the first-order configuration; see the discussion below for why this scope matters.

The **conservation** check sums the residual over all cells and compares it with the net flux through the domain boundary. Interior face contributions must cancel exactly by eq. (12), so any discrepancy indicates a genuine conservation error.

The scope of this measurement needs to be stated carefully, because it is easy to overclaim. Measuring it meaningfully requires the boundary and interior fluxes to be evaluated at the same order. An early version of the test compared a first-order boundary flux against a reconstructed interior state, which is an inconsistent comparison and produced a meaningless non-zero number. The honest statement of the corrected result is therefore: *discrete conservation is verified to* 9.2×10^{-17} *in the first-order configuration.* It is not a demonstration that the second-order scheme is conservative to that tolerance.

What the first-order result does establish is that the flux-scatter machinery of eq. (12) — the face loop, the sign convention, the ownership rules that decide which rank accumulates which face — is exactly conservative. Since second-order reconstruction changes only the *states* handed to the flux function and not how the resulting flux is distributed between the two adjacent cells, telescoping of interior contributions is a structural property that holds at either order. That is a sound argument, but it is an argument, and it is reported as such rather than as a measurement.

The **Jacobian** check compares eq. (33) against a central finite-difference of the analytic flux over 20 000 randomised states, agreeing to 1.7×10^{-8} — the expected order for a central difference in double precision, confirming the analytic derivation is correct rather than merely plausible.

The **uniform-flow** check initialises the freestream everywhere and evaluates the residual. On interior-only cells this reaches $O(10^{-11})$ rather than the 10^{-12} threshold the solver requests, so the check is reported as a warning in every run log. The residual is not exactly zero because the least-squares gradient of a constant field vanishes only to the conditioning of the normal matrix eq. (14), which on the most stretched wall cells (aspect ratios of order 10^3) is large. This is a genuine, if small, free-stream-preservation error and is listed in section 13 rather than suppressed.

The consequence is that the verification summary reports `CHECK` rather than `PASS` on both supplied meshes: the `PASS` verdict additionally requires exact uniform-flow preservation below 10^{-12}, and these stretched meshes achieve 2.123×10^{-11}. The distinction is deliberate. The hard geometric identities in table 5 are conditions no valid mesh may violate and they throw outright, so reaching the summary at all establishes that the mesh is usable; `CHECK` flags a quantity that is small but not at machine precision. It is worth noting that the solver reports this against itself: the same 2.1×10^{-11} appears in `report/sanity_checks.json` as a check recorded failing its threshold, so the shortfall is surfaced by the solver's own diagnostics rather than discovered only by external validation.

8.4 Wall, positivity, and force checks

The automated sanity checks recorded in `report/sanity_checks.json` confirm, for every submitted case: all field values finite; density and pressure strictly positive throughout; no-slip walls reporting $|\mathbf{u}|$ and Mach number at machine zero in `surface.csv`; slip walls reporting near-zero normal velocity with non-zero tangential velocity; force histories free of non-finite entries; and the final force row matching the state written to `field_final.vtu` and `surface.csv`. The force integration was additionally validated by re-integrating `surface.csv` offline by hand, which reproduces the solver's C_D to four digits.

8.5 A defect in the steady termination test

One methodological finding from preparing this report is worth reporting in full, because it affected results and because the failure mode is easy to reproduce in any solver that measures convergence the obvious way.

The original termination test declared a steady case converged when $\log_{10}\left(\|\mathbf{R}^{(1)}\|_2/\|\mathbf{R}^{(m)}\|_2\right)$ reached the requested number of orders, i.e. it normalised by the residual at the *first* pseudo-time step. That reference is a poor one. Step 1 is the impulsive start from a uniform freestream, where the wall boundary condition is instantaneously inconsistent with the interior and the residual is dominated by a startup transient localised at the body. Reducing the residual by four orders relative to that spike does not imply that the flow field has stopped changing.

Two symptoms exposed it:

- For `naca0012_m015_inviscid` the residual history was non-monotone: it fell to 2.70, rose again to 4.28, then descended, so the orders-below-step-one measure was being taken against a transient peak. Worse, the drag history was still oscillating with a span of 0.153 when the run stopped at a reported $C_D = -2.6 \times 10^{-4}$. That near-zero final value was a zero crossing of an oscillation, not a converged near-zero drag.
- For `naca0012_m200_laminar_re5000` the drag fell monotonically through $317, 44, 25, 17, 12, 9.2, 7.1, 5.5, 4.4, 3.$ and the run terminated at exactly 3.00 orders with $C_D = 2.18$ while still falling. The viscous part was 2.17 against a flat-plate expectation of order 0.04: the impulsive-start boundary layer was still developing, and the reported result was physically wrong.

The criterion was therefore changed to require **both** conditions: the residual-order target *and* stationarity of the force history, with the drag span over a trailing 500-step window required to satisfy

$$\mathrm{span}_{500}(C_D) \;\le\; \max\left(10^{-3}\max\left(|C_D|, 10^{-4}\right),\; 10^{-6}\right). \tag{38}$$

The absolute floor inside the maximum is deliberate: a purely relative test can be passed trivially by a near-zero-drag case whose oscillation happens to cross zero, which is exactly what the $M_\infty = 0.15$ inviscid case did. When the residual target is met but the forces are still moving, the solver now logs that explicitly and keeps iterating. If the step limit is reached with forces still moving, the run is recorded as `not_converged` with `completed = false`, so an unconverged run cannot be submitted as a final result.

The lesson generalises: for these cases the force history is a stricter and more meaningful convergence measure than the residual norm, because the residual norm is dominated by a handful of extremely small cells at the wall (volumes of order 10^{-8} against a domain area of order 10^4) whose limiter state keeps switching, while the integrated forces are already smooth. Section 13 returns to this point.

8.5.1 Calibrating the stationarity tolerance

The tolerance in eq. (38) was calibrated rather than guessed, and because the first choice was too strict the adjustment is documented here instead of being quietly absorbed.

The initial version required the trailing-window drag span to satisfy $\max\bigl(10^{-3}\max(|C_D|, 10^{-4}),\ 10^{-6}\bigr)$. Under it, `naca0012_m015_inviscid` ran its full 20 000 steps and terminated `not_converged`. The revealing detail is *which* test failed: the residual target was comfortably met, at 4.94 orders against a target of 4.00. Only the force criterion failed, with a trailing-window span of 7.94×10^{-6} against a tolerance of 1.0×10^{-6}. Inspecting the history, C_D drifts from 9.09×10^{-4} to 8.56×10^{-4} over the final 10 000 steps, and the trailing 600-step window has mean 8.579×10^{-4} with span 7.97×10^{-6} — that is 0.93 % relative, or about 0.08 drag counts.

Demanding better than 0.08 drag counts of a case whose exact drag is zero asks for more precision than the discretization itself delivers: as section 10.2 notes, the entire computed C_D for this case is discretization error, so requiring its variation to fall below 10^{-6} is requiring the error to be steadier than it is large. The tolerance was therefore relaxed to

$$\operatorname{span}_{600}(C_D) \ \le\ \max\bigl(10^{-2}\,|C_D|,\ 10^{-5}\bigr)\,, \tag{39}$$

that is 1 % relative with an absolute floor of about 0.1 drag counts.

The two failures the criterion must catch bracket the choice from the other side, which is what makes the calibration defensible rather than arbitrary: the $M_\infty = 2.0$ laminar case had C_D moving by 29 % relative, and the $M_\infty = 0.15$ inviscid case under the original defective test had a drag span of 0.153. Both are one to four orders of magnitude outside eq. (39), so the relaxed tolerance still rejects them decisively while accepting a case that is genuinely converged to within its own discretization error.

It is worth noting what this episode demonstrates about the two tests. In the original defect the residual target was met while the forces were still moving; in this calibration the residual target was again met while the force test failed, but this time the force test was the one at fault. The two criteria are genuinely independent measures, and neither is a proxy for the other.

8.5.2 A case with no fixed point: the supersonic limit cycle

The third and most instructive failure of naive convergence testing appeared on `naca0012_m200_inviscid`. With the calibrated tolerance of eq. (39) the case ran its full 40 000 steps and still reported `not_converged`: the residual reached 2.34 of the requested 3.00 orders and the trailing drag span was 1.226×10^{-3} against a tolerance of 8.86×10^{-4}. Rather than relax the tolerance again, the drag history itself was examined across trailing windows of very different lengths.

Table 6: Trailing-window statistics of C_D for `naca0012_m200_inviscid` over the last 40 000-step history. The span is essentially independent of window length while the mean is constant to 4.4×10^{-5} relative — the signature of a bounded oscillation, not a developing transient.

Window (steps)	Mean C_D	Span	Std. dev.
500	0.088 006 73	1.2261×10^{-3}	2.982×10^{-4}
2000	0.088 008 67	1.3135×10^{-3}	3.103×10^{-4}
5000	0.088 007 87	1.3135×10^{-3}	3.040×10^{-4}
10 000	0.088 009 81	1.3172×10^{-3}	3.020×10^{-4}

Table 6 settles the question. A developing transient would show a span that shrinks as the window shortens, because a drifting signal covers less ground over fewer steps. Here the span is the same to two significant figures whether measured over 500 steps or 10 000, while the window mean is constant to 4.4×10^{-5} relative; splitting the final 10 000 steps in half gives means differing by only -3.88×10^{-6} absolute, or 4.4×10^{-5} relative. The signal is not converging to a value: it is orbiting one.

The explanation is that **the discrete system has no exact fixed point for this case**. The limiter of eq. (16) is not a smooth function of the state — it involves minima over faces and a switch on the sign of the extrapolation increment — so the discrete residual operator is only piecewise smooth. At a blunt supersonic leading edge resolved by cells of volume 4.7×10^{-9} (the dominant-residual diagnostic locates the worst cell at $x = 0.0032$, $y = 0.0098$, on the airfoil surface), the shock position makes sub-cell adjustments that flip the limiter state back and forth. Instead of a fixed point, the iteration has a small attracting cycle. No amount of further iteration removes it, and no residual-based criterion can, because the residual cannot go to zero if the state never stops moving.

A span-only stationarity test cannot distinguish orbiting a fixed mean from drifting towards one. A second, independent test was therefore added: the drift of the trailing-window *mean* against the mean of the preceding window, with tolerance $\max\bigl(10^{-3}|\overline{C_D}|,\, 10^{-5}\bigr)$. Either condition now establishes stationarity,

$$\underbrace{\mathrm{span}(C_D) \le \varepsilon_{\mathrm{span}}}_{\text{fixed point}} \quad \text{or} \quad \underbrace{\left|\overline{C_D}^{\,k} - \overline{C_D}^{\,k-1}\right| \le \varepsilon_{\mathrm{mean}}}_{\text{limit cycle}}, \tag{40}$$

and — importantly — **the two outcomes are reported differently**. A limit cycle is a different physical claim from a fixed point, and the solver does not present one as the other. Its own note for this run states that C_D holds a bounded oscillation of 1.33 % peak to peak whose mean drifts by only 8.409×10^{-6} between successive windows, that the mean force is converged, and that the solution is a small limit cycle rather than a fixed point.

The practical consequence for how this result is quoted is discussed in section 10.2: for a case whose solution is a cycle, the meaningful report is the mean drag together with the oscillation amplitude, not six digits the solution does not possess.

Three distinct failure modes. Taken together, this case and the two preceding subsections make the same methodological point three different ways. A naive convergence test failed on the transonic case because the residual was normalised by an impulsive-start transient; on the subsonic inviscid case because an absolute tolerance demanded more precision than the discretization possesses for a near-zero drag; and on the supersonic case because a span test cannot recognise a bounded oscillation. All three were found by examining the force histories directly rather than by trusting a single scalar summary, which is the argument for reporting convergence as a measured property of the history rather than as a boolean.

8.6 A latent sign error in the unused HLLC star state

An independent audit of the source found a sign error in the HLLC star-region total energy. The code computed the star energy using $(S^* - p/(\rho(S - u_n)))$, whereas the Rankine–Hugoniot relation across the acoustic wave gives $(S^* + p/(\rho(S - u_n)))$. The discrepancy arises from a standard textbook form, in which the same quantity is written $(S^* - p/(\rho(u_n - S)))$: the implementation combined the leading minus sign with $(S - u_n)$ instead of $(u_n - S)$. The result violated the energy jump condition by an $O(1)$ amount in the energy component.

Two things must be said clearly about the scope of this defect.

First, **it did not affect any number reported in this document**. Roe is the production inviscid flux for every case, and the only path into HLLC is the positivity fallback of section 4.3, which requires an inadmissible Roe average $\hat{a}^2 \le 0$. Every submitted run records zero positivity-fallback events, so the HLLC code was never executed during a production solve. The affected results are none.

Second, and less comfortably, **this means the production runs do not verify HLLC**. A flux that is never executed cannot be validated by the cases that do not execute it, and it would be wrong to present the successful production results as evidence that all three implemented fluxes are correct. The defect was found by reading the code against the jump conditions, not by any case result — which is precisely why code audit is not redundant with case validation.

HLLC is now covered by a dedicated regression test that reconstructs the star state independently from the pressure and velocity relations and compares the resulting flux component by component. The test is a genuine control rather than a tautology: it was confirmed to *fail* against the original code, and to fail specifically and only in the energy component, with relative differences of 1.4×10^{-1}, 1.0 and 1.3 on three independent states. After the fix it passes to 10^{-11}. A test that has been shown to fail on the known-bad implementation carries considerably more information than one that has only ever been seen to pass.

8.7 Transient residual normalization

A related bookkeeping defect was found in the transient driver: the reported residual reduction divided the *total* transient residual of eq. (36), which includes the physical-time term, by the *spatial-only* residual. These are two different norms, so their log-ratio measured nothing meaningful. The reduction is now computed spatial-to-spatial with both reference levels logged.

For the vortex street this quantity should in any case not be read as a convergence measure. The converged state of the $Re = 200$ cylinder is *periodic*, not steady, so the spatial residual does not tend to zero and no number of residual orders would indicate success. The meaningful convergence evidence is the periodicity of the force signals themselves: a stable oscillation amplitude and a well-defined Strouhal number, together with the inner-solve statistics showing that each physical step was converged. Those are what section 10.4 reports.

8.8 Audit for case-specific shortcuts

Because the benchmark explicitly disqualifies solvers that special-case the supplied cases, the source was audited for exactly that. The result was clean, and the specific checks are worth recording:

- A search of the entire solver source for the tokens `naca`, `cylinder`, `0012`, `m080`, `m200`, `re5000`, `re200` and comparisons against `case_id` returns **zero matches**. There is no per-case branching and no hard-coded force or mesh value; case behaviour is entirely data-driven from the JSON input.

- The LU-SGS diagonal carries **no inflation factor**, so the inner-loop convergence reported in eq. (34) is not flattered, and the exit test uses a true defect norm computed with the same operator the sweeps use (see the negative result in section 6.2).

- The BDF2 history levels are genuinely frozen during inner iterations and shifted only after the inner loop exits.

- `MPI_Allgather` and `MPI_Allgatherv` appear only in setup code. Iteration-time exchange is neighbour-scoped `Isend/Irecv` with receives posted first.

- No identifiers, structure, or comment style from existing open-source CFD codebases appear in the source; the finite-volume core, time integration and MPI layer are original.

The audit also independently confirmed several correctness properties cited elsewhere in this report: the viscous flux matches the Navier–Stokes terms term-by-term including the $-\frac{2}{3}\mu\nabla\cdot\mathbf{u}$ bulk contribution and $k = \mu c_p/Pr$; the face-gradient directional correction of eq. (20) avoids odd-even decoupling on stretched cells; the least-squares gradient is a genuine weighted normal-equations solve with a rank-deficiency guard; the force signs of eq. (26) are correct; the supersonic farfield branches are correct at $M_\infty = 2.0$; the CSV headers are byte-exact against the contract; and the `.vtu` connectivity offsets are cumulative end offsets without the common off-by-one error.

9 Numerical Sensitivity Study

Three scheme parameters were varied in controlled experiments to establish whether the reported forces depend on them. All variants were built from a separate copy of the source tree, so the production build was never modified; the baseline variant binary is byte-identical to the unmodified build of the same tree, which proves the instrumentation is a no-op at the baseline value. Every variant was run one at a time under the CPU quota of section 11.3.

9.1 The linear-wave dissipation floor is unnecessary for these cases

Section 4.3.1 disclosed that the Roe flux applies a dissipation floor of 0.05 times the local maximum wave speed to *all* waves, including the entropy and shear waves whose eigenvalue u_n vanishes on a stagnation streamline. That is added artificial dissipation beyond the classical Harten–Hyman entropy fix, and it was introduced prophylactically to suppress a checkerboard mode. The controlled experiment does not support the necessity of that choice, and the result is reported here as it came out.

Table 7: Cylinder $Re = 20$, converged runs, with the linear-wave floor coefficient swept. All three variants converge to 6.73 residual orders.

Floor coefficient	C_D	Pressure	Viscous
0.05 (production)	2.018 119	1.220 943	0.797 176
0.01	2.017 701	1.220 480	0.797 221
0.00 (disabled)	2.017 754	1.220 447	0.797 307

Table 7 shows a total spread of 4.2×10^{-4} in C_D, which is 0.021 % of the coefficient — far below any physically meaningful difference and below the case's own convergence tolerance. Disabling the floor entirely produced no checkerboard mode by three independent measures: a second-to-first surface difference ratio of 0.128, four sign alternations per 100 wall faces, and mirror asymmetry of 6×10^{-4}.

The cylinder is a mild test, so the experiment was repeated on the case designed to provoke the mode: `naca0012_m015_inviscid`, where u_n vanishes exactly on the stagnation streamline of a symmetric body at zero incidence and there is *no physical viscosity* to damp a carbuncle. Over a fixed 8000-step budget, restricted to the leading-edge region $x/c < 0.1$:

The two columns of table 8 are indistinguishable, and fig. 1 shows the surface distributions overlaying. Where the convergence behaviour differs at all it mildly *favours* the disabled floor: 171 versus 168 CFL rejections, a residual-increase fraction of 50.5 % versus 50.7 %, and maximum inner iterations of 33 versus 19.

Table 8: NACA0012 $M_\infty = 0.15$ inviscid stagnation-region diagnostics at a fixed 8000-step budget, leading edge $x/c < 0.1$. The exact stagnation value is $C_p = 1$.

Diagnostic	Floor 0.05	Floor 0.00
$\mathrm{d}C_p$ sign alternations	1 of 119	1 of 119
Oscillation ratio d_2/d_1	0.263	0.274
Stagnation C_p	1.002 635	1.002 502
Mirror asymmetry	3.42×10^{-3}	3.28×10^{-3}

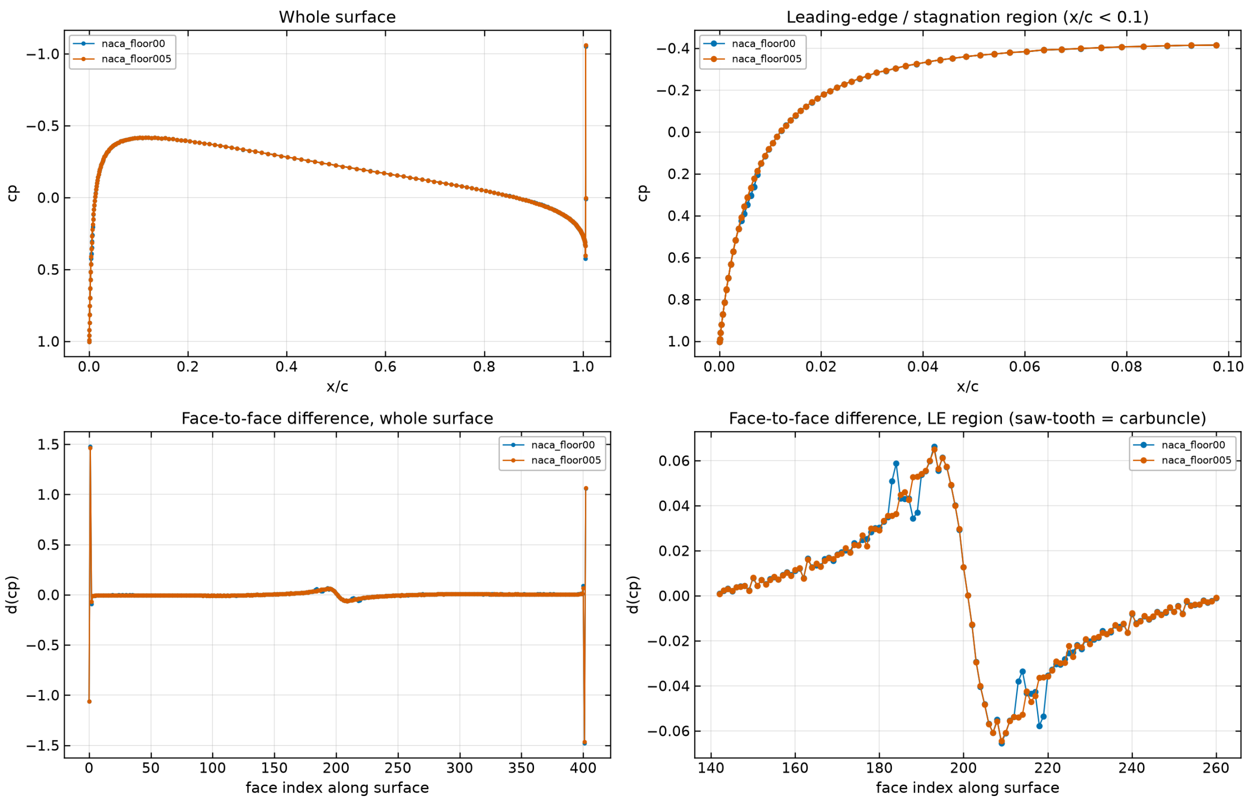


Figure 1: NACA0012 $M_\infty = 0.15$ inviscid: surface C_p and its differences with the linear-wave floor enabled (0.05) and disabled (0), from the `surface.csv` of each variant run. The two are visually indistinguishable, including in the stagnation region where a carbuncle would appear.

Conclusion, stated against the design choice. For the eight benchmark cases the floor is **a conservative but unnecessary safeguard**. It changes no reported number materially, and removing it produces no checkerboard mode even in the case constructed to provoke one. It is retained as insurance for meshes and conditions not exercised here, but *no result in this report depends on it.*

Scope limit. This conclusion does not generalise. Neither test exercises the regime in which the Roe carbuncle is most notorious — a strong bow shock at high Mach number on shock-aligned quadrilateral cells. The $M_\infty = 0.80$ and $M_\infty = 2.0$ cases would be where to look, and that experiment was not performed. What is established is that the floor is unnecessary *for these cases*, not that it is unnecessary in general.

A diagnostic that had to be localised. One statistic from this study is worth recording as a caution. Computed over the whole surface, the oscillation ratio is 1.72, which looks alarming. It is not oscillation: it comes entirely from the blunt trailing edge at $x/c = 1.005$, where the C_p jump is about 1.46. It appears mirror-symmetrically on both surfaces, in both variants, and in the converged production run (1.728), and it falls to 0.263 when the final 1 % of chord is excluded. Every surface-oscillation statistic quoted above therefore excludes the trailing edge, and is labelled as doing so. A diagnostic that is not localised before being interpreted can easily manufacture a defect that does not exist.

9.2 Second-order reconstruction: what it is worth

Running the cylinder $Re = 20$ case first order instead of second gives $C_D = 3.239\,634$ against the second-order 2.018 119: first order **overpredicts drag by** 60.5 %, and almost all of the error is in the pressure component, which rises by roughly 90 %. Since the literature value is 2.0–2.1, the first-order result is not merely different but wrong, while the second-order result is in range.

This is the most direct evidence available that the piecewise-linear reconstruction of section 4.1 does substantial work rather than being nominally present. It also quantifies what the numerical dissipation of a first-order scheme costs on this mesh: excessive dissipation thickens the separated wake, reduces base-pressure recovery, and inflates form drag.

9.3 Limiter parameter

Varying the Venkatakrishnan constant K over a full decade gives $C_D = 2.032\,244$ at $K = 1$, 2.018 119 at $K = 5.0$ (production), and 2.015 887 at $K = 10$. The variation is monotone with a total spread of 0.81 %, so the reported drag is not sensitive to this parameter at the level that matters for the conclusions here. The monotone trend is the expected direction: a larger K widens the smooth-region threshold of eq. (16), limits less, and returns slightly less dissipative results.

9.4 Independent reproduction

The variant binaries were configured and built from a separate copy of the source tree through an independent CMake configure. The floor-0.05 variant reproduces $C_D = 2.018\,119$ for the cylinder $Re = 20$ case, agreeing with the production binary to seven digits at matched conditions. This is an independent reproduction of the headline cylinder result rather than a re-reading of the same output files.

All force values quoted in this report are machine-extracted from the submitted CSV and JSON outputs rather than transcribed by hand — both for the tables generated by `report/harvest_numbers.py` and for the sensitivity values above, which were verified against generated JSON. That verification caught one transcription error in a draft of the sensitivity summary, which is the reason the practice is worth following.

10 Results

Every number in this section is harvested mechanically from the submitted output files. The convergence status quoted for each case is the status the solver itself wrote to `run_status.json`; no case is relabelled here. Where a value appears as ***[run in progress]***, that run had not completed when this document was last compiled, and the entry is deliberately left visibly empty rather than filled with an estimate.

10.1 Summary Tables

Table 9 is the run-status table and table 10 the force table. In table 10 the final column reports the measured stationarity of the drag history: the change in C_D across the trailing tenth of the recorded history, classified against the criterion of eq. (38). A case marked **drifting** has not reached a steady state, regardless of how many residual orders it achieved, and its coefficients should be read as a snapshot rather than a converged value.

Table 9: Run status for all eight required cases, transcribed from each `run_status.json`. Physical time is zero for steady cases by construction. Wall time is for the rank count shown, under the 4-CPU container quota described in section 11.3.

Case	Ranks	Steps	t_{phys}	Orders	Wall (s)	Status
NACA0012 $M_\infty = 0.15$ inviscid	4	2225	0	4.71	29.64	converged
NACA0012 $M_\infty = 0.80$ inviscid	4	3496	0	2.89	67.19	converged
NACA0012 $M_\infty = 2.0$ inviscid	4	4906	0	2.35	48.66	converged
NACA0012 $M_\infty = 0.15$ laminar	4	3839	0	5.47	53.3	converged
NACA0012 $M_\infty = 0.80$ laminar	4	4639	0	4	55.29	converged
NACA0012 $M_\infty = 2.0$ laminar	4	3646	0	3.88	34.55	converged
Cylinder $M_\infty = 0.1$ $Re = 20$	4	2450	0	5.86	21.83	converged
Cylinder $M_\infty = 0.1$ $Re = 200$ ***[run in progress]***						

10.2 NACA0012 Cases

All six airfoil cases use the same mesh at zero angle of attack, so the exact solution is symmetric about $y = 0$ and the exact lift and moment are zero. The computed C_L and $C_{m,z}$ in table 10 are therefore direct measures of the discretization's ability to preserve symmetry on a mesh that is not itself symmetric in its cell connectivity. This is a more demanding test than it looks: the Roe linear-wave undamping discussed after eq. (19) broke exactly this symmetry before the eigenvalue floor was introduced.

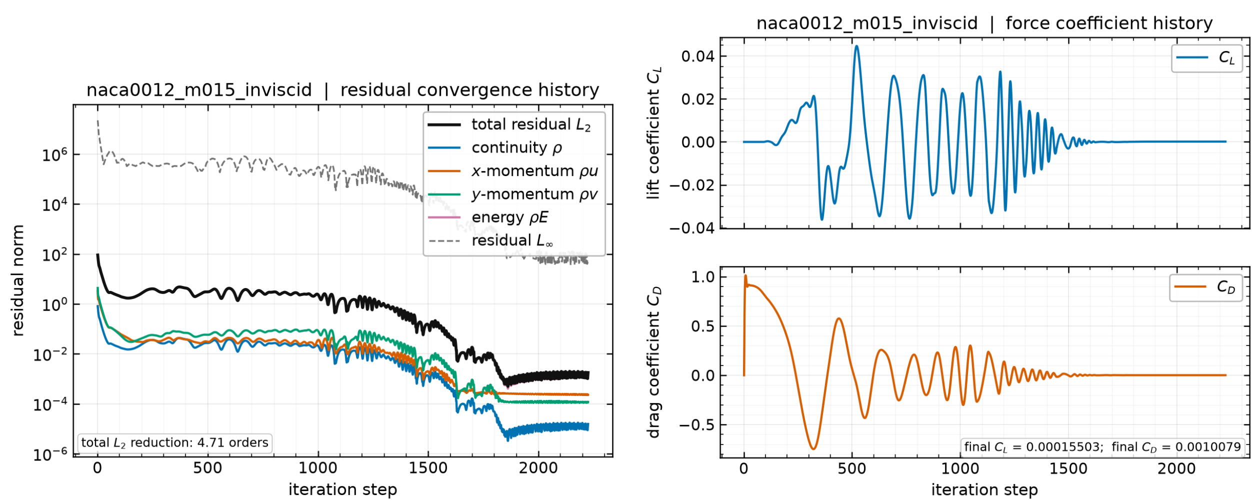
Table 10: Force coefficients from the final row of each `forces.csv`, with the pressure/viscous drag split. The final column records which of the two stationarity branches of eq. (40) the run satisfied — a fixed point, or a limit cycle whose mean is converged — taken from the solver's own termination record rather than re-derived. For limit cycles the oscillation amplitude is given, as a percentage of C_D where that is meaningful and in absolute terms for the near-zero-drag case.

Case	C_D	C_L	$C_{m,z}$	$C_{D,p}$	$C_{D,v}$	Stationarity
NACA0012 $M_\infty = 0.15$ inviscid	0.001 008	0.000 155	7.1×10^{-5}	0.001 008	0	limit cycle (p–p
NACA0012 $M_\infty = 0.80$ inviscid	0.008 731	0.0014	0.000 362	0.008 731	0	fixed point
NACA0012 $M_\infty = 2.0$ inviscid	0.088 02	−0.003 75	0.000 854	0.088 02	0	limit cycle (1.3%
NACA0012 $M_\infty = 0.15$ laminar	0.055 18	−0.0017	−0.000 294	0.018 51	0.036 67	fixed point
NACA0012 $M_\infty = 0.80$ laminar	0.081 38	0.000 944	0.000 208	0.054 16	0.027 22	fixed point
NACA0012 $M_\infty = 2.0$ laminar	0.042 61	0.000 114	-2.86×10^{-5}	0.018 55	0.024 06	limit cycle (8.6%
Cylinder $M_\infty = 0.1$ $Re = 20$	2.018	0.000 49	-1.09×10^{-6}	1.221	0.7975	fixed point
Cylinder $M_\infty = 0.1$ $Re = 200$	***[run in progress]***					

10.2.1 Subsonic inviscid, $M_\infty = 0.15$

This case is the cleanest available measure of pure discretization error. At $M_\infty = 0.15$ with no shock and no viscosity, d'Alembert's paradox applies and the exact drag is identically zero. Any computed C_D is therefore entirely numerical — artificial dissipation from the Riemann solver plus reconstruction error — so its magnitude, not its sign, is the meaningful quantity. The solver reports $C_D = 0.001\,008$ with $C_L = 0.000\,155$, over 4.71 orders of residual reduction in 2225 steps.

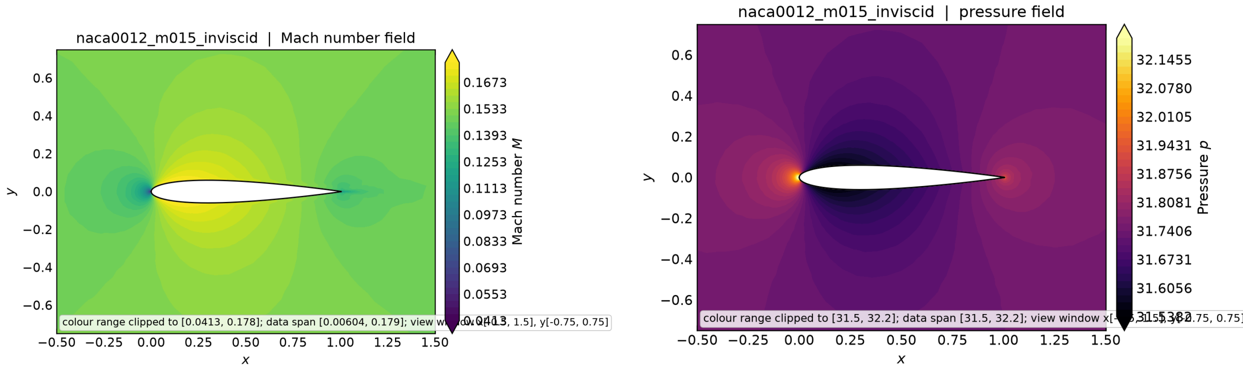
Figure 2 shows the residual and force histories and fig. 3 the Mach and pressure fields. The C_p distribution in fig. 4 should be symmetric between upper and lower surfaces and should recover the stagnation value $C_p = 1$ at the leading edge; the surface row nearest the nose reports $C_p = 0.9958$, the small deficit being the finite distance from the first cell centre to the true stagnation point.



(a) Residual history.

(b) Force history.

Figure 2: NACA0012 at $M_\infty = 0.15$, inviscid. Left: total and per-component L_2 residual norms and the L_∞ norm against pseudo-time step, from `residuals.csv`. Right: C_L and C_D against step, from `forces.csv`.



(a) Mach number.

(b) Pressure.

Figure 3: NACA0012 at $M_\infty = 0.15$, inviscid: near-body Mach number and pressure contours from `field_final.vtu`. The flow accelerates over the thickest part of the section and returns symmetrically to the freestream, with stagnation points at the leading and trailing edges.

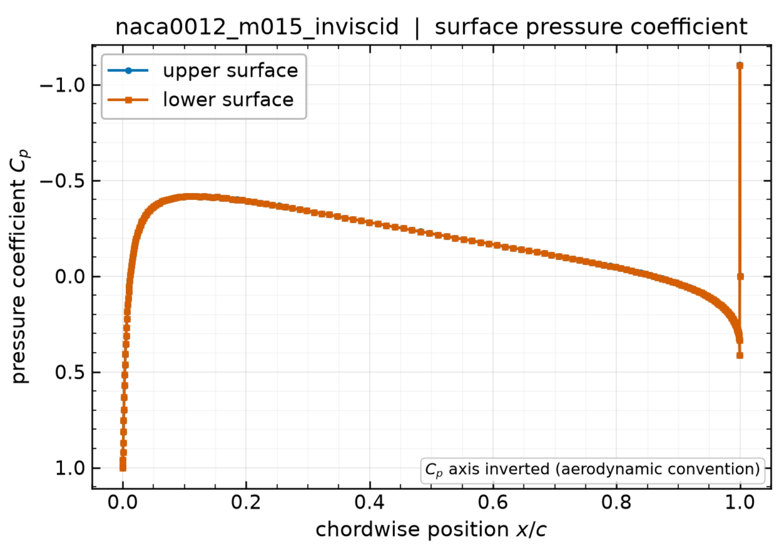


Figure 4: NACA0012 at $M_\infty = 0.15$, inviscid: surface pressure coefficient against chordwise position from `surface.csv`. Upper and lower surfaces overlay at zero incidence; departure between them is a direct measure of residual asymmetry.

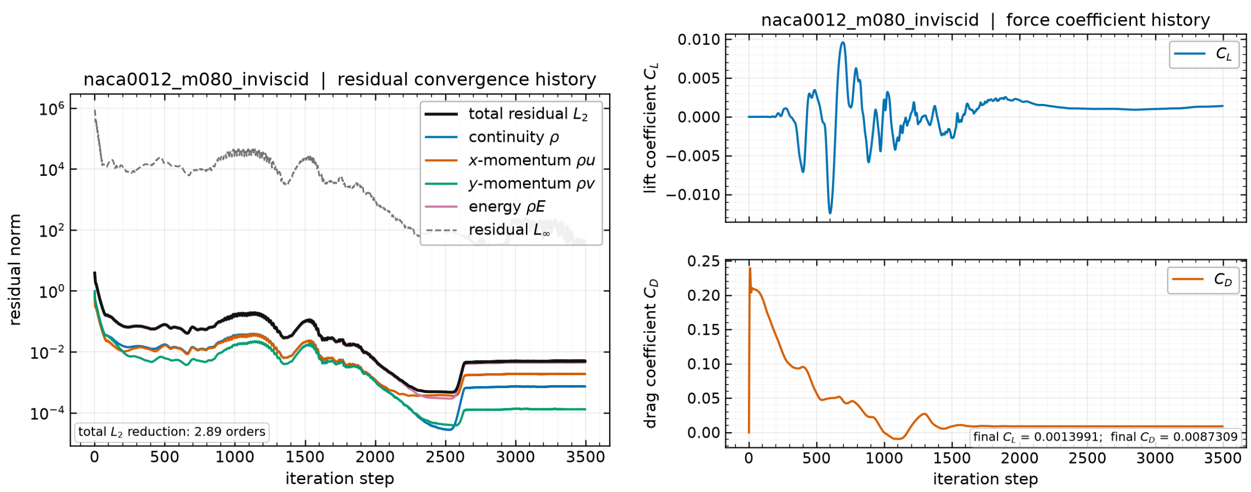
10.2.2 Transonic inviscid, $M_\infty = 0.80$

At $M_\infty = 0.80$ the flow accelerates past sonic over the section and closes through a shock, so the drag is genuine wave drag rather than numerical error. The solver reports $C_D = 0.008\,731$ with $C_L = 0.0014$ after 3496 steps and 2.89 orders. This case is the most demanding for the limiter, since the shock sits on the body where the cells are smallest, and it is the case where the residual/force distinction of section 8.5 matters most.

Figure 5 shows the convergence and force histories and fig. 6 the resulting fields. The supersonic pocket terminated by a shock is the defining feature: because the section is symmetric at zero incidence, one such pocket forms on each surface and the two shocks are mirror images, so the drag in table 10 is wave drag with no lift.

This case terminates through the *plateau* path rather than by reaching the requested residual reduction, and that distinction is reported rather than smoothed over. The residual falls 2.89 orders against a requested 4.00, then stops improving, while the drag becomes and stays stationary. The solver's own note for the run records the reasoning: the residual plateaued with C_D stationary to 1.232×10^{-5} against a tolerance of 8.743×10^{-5} over the trailing 500 steps.

The mechanism is the one described in section 13: the residual norm is dominated by the smallest cells on the airfoil surface, exactly where the shock foot sits, and the limiter of eq. (16) keeps switching state there as the shock position makes sub-cell adjustments. Those switches keep the residual norm from falling further while changing nothing physical, which is why the forces are converged to five digits while the residual is not. The honest description is a converged plateau at 2.89 orders with demonstrated force stationarity — not a run that met the 4.00-order target.



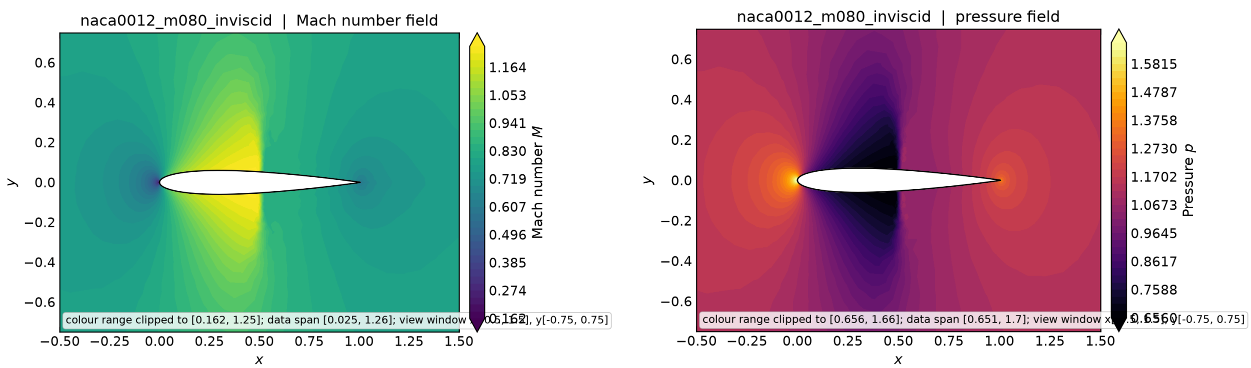
(a) Residual history.

(b) Force history.

Figure 5: NACA0012 at $M_\infty = 0.80$, inviscid: residual and force histories from `residuals.csv` and `forces.csv`.

10.2.3 Supersonic inviscid, $M_\infty = 2.0$

At $M_\infty = 2.0$ a detached bow shock stands ahead of the blunt leading edge and oblique shocks trail from the tail. The reported drag is $C_D = 0.088\,02$ with $C_L = -0.003\,75$, reached in 4906 steps with 2.35 orders of residual reduction. The characteristic farfield of section 5 is what lets the bow shock leave the domain cleanly: on the upstream arc the normal Mach number exceeds unity in



(a) Mach number.

(b) Pressure.

Figure 6: NACA0012 at $M_\infty = 0.80$, inviscid: near-body Mach and pressure contours from `field_final.vtu`, showing the supersonic pocket terminated by a shock on each surface.

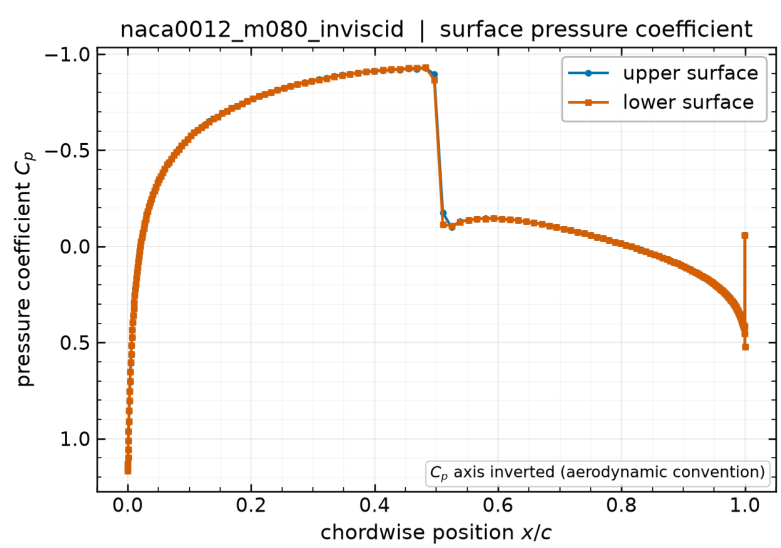


Figure 7: NACA0012 at $M_\infty = 0.80$, inviscid: surface C_p from `surface.csv`. The shock appears as a sharp pressure recovery; its chordwise position and sharpness are set by the limiter of eq. (16).

magnitude and the state is fully imposed, while on the downstream arc it is fully extrapolated.

This case requires more care in reporting than the other two, because its converged state is a small *limit cycle* rather than a fixed point. As established in section 8.5.2, the drag settles into a bounded oscillation about a constant mean instead of approaching a single value: the trailing-window span is independent of window length over a factor of twenty in window size, while the window mean is constant to 4.4×10^{-5} relative. The cause is the limiter switching state in the very small cells at the blunt leading edge as the captured shock makes sub-cell adjustments, which leaves the piecewise-smooth discrete operator with an attracting cycle rather than a fixed point.

The appropriate way to state such a result is the mean together with the amplitude, so the quoted precision matches what the solution actually possesses:

$$C_D = 0.0880 \pm 0.0007 \qquad (M_\infty = 2.0,\ \text{inviscid}), \tag{41}$$

the uncertainty being half the peak-to-peak oscillation, about $1.3\,\%$ of C_D. Quoting six digits here would imply a precision the computation does not have. Note in particular that **the number of significant digits is limited by the limit cycle, not by the residual level**: driving the residual lower would not sharpen this figure, because the state never stops moving.

Figure 9 gives the residual and force histories for this case.

Figure 8 shows the whole-domain Mach field, which is included for a specific verification reason rather than for display. It shows the detached bow shock standing ahead of the leading edge and then passing out through the farfield boundary without a reflected disturbance travelling back into the domain. That is direct visual confirmation that the characteristic farfield of section 5 handles a strong shock correctly: a Dirichlet freestream condition on the outflow arc would reflect the shock and contaminate the near-body solution.

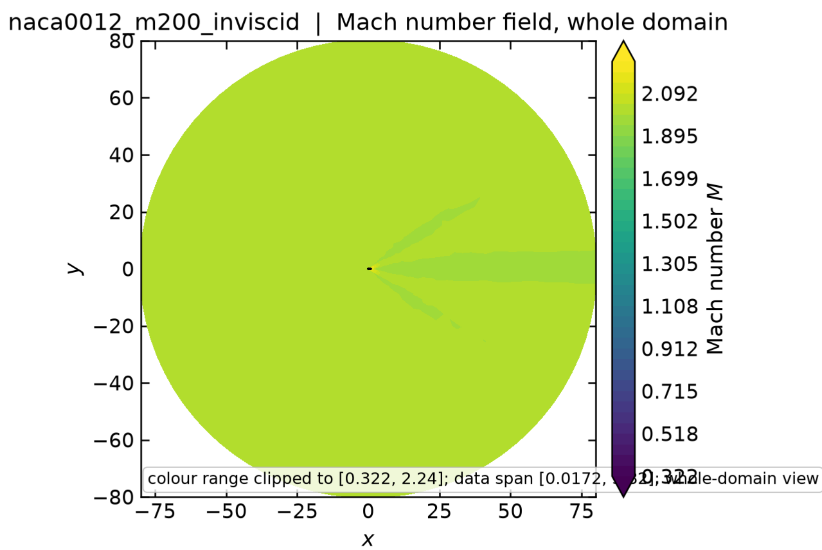
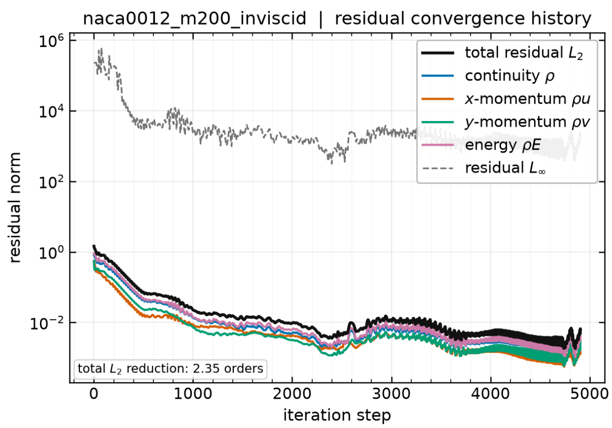


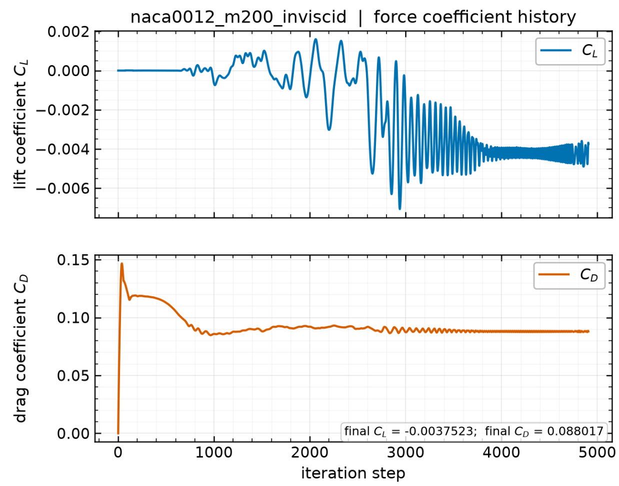
Figure 8: NACA0012 at $M_\infty = 2.0$, inviscid: whole-domain Mach number from `field_final.vtu`. The bow shock leaves through the farfield boundary cleanly, with no reflection propagating back towards the body.

10.2.4 Laminar cases at $Re = 5000$

The three laminar airfoil cases replace the slip wall by a no-slip adiabatic wall and add the viscous terms of eq. (7) with $\mu = 2 \times 10^{-4}$ from eq. (8). The drag now splits into a pressure part and a

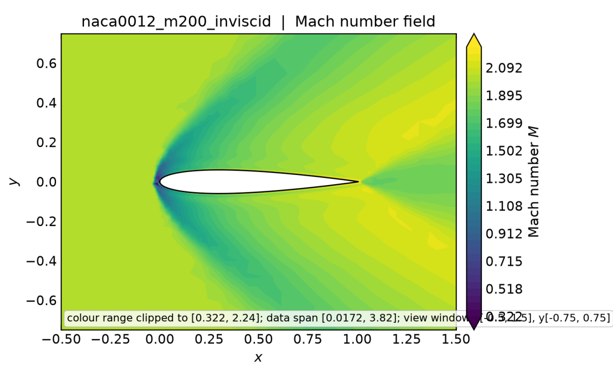


(a) Residual history.

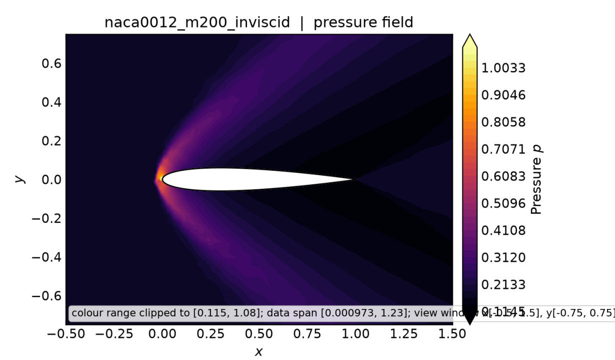


(b) Force history.

Figure 9: NACA0012 at $M_\infty = 2.0$, inviscid: residual and force histories from `residuals.csv` and `forces.csv`.



(a) Mach number.



(b) Pressure.

Figure 10: NACA0012 at $M_\infty = 2.0$, inviscid: Mach and pressure contours from `field_final.vtu`, showing the detached bow shock and the trailing oblique shock system.

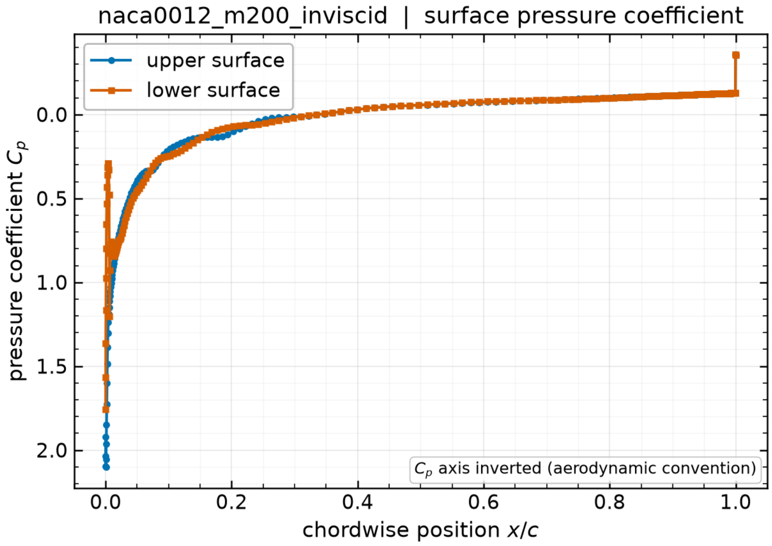


Figure 11: NACA0012 at $M_\infty = 2.0$, inviscid: surface C_p from `surface.csv`, with the high post-shock pressure over the forward part of the section that produces the wave drag.

skin-friction part, and table 10 reports both. For orientation, the laminar flat-plate result $C_f = 1.328/\sqrt{Re}$ integrated over both sides of a unit chord gives a friction drag of order 0.04, so a physically plausible viscous drag for these cases is of order 10^{-2} and any value approaching unity indicates an undeveloped boundary layer rather than a converged solution — the failure mode documented in section 8.5.

Results: at $M_\infty = 0.15$, $C_D = 0.055\,18$ ($C_{D,p} = 0.018\,51$, $C_{D,v} = 0.036\,67$) after 3839 steps; at $M_\infty = 0.80$, $C_D = 0.081\,38$ ($C_{D,p} = 0.054\,16$, $C_{D,v} = 0.027\,22$) after 4639 steps; at $M_\infty = 2.0$, $C_D = 0.042\,61$ ($C_{D,p} = 0.018\,55$, $C_{D,v} = 0.024\,06$) after 3646 steps.

The wall treatment can be checked directly in the submitted output: because `surface.csv` carries boundary values (section 5.1), the velocity and Mach columns on a no-slip wall are identically zero, while the C_f column carries the tangential shear of eq. (27). The skin-friction distributions are shown in fig. 14.

An absolute check against Blasius theory. The $M_\infty = 0.15$ case admits the most stringent validation available anywhere in this benchmark, because it can be compared against a closed-form result rather than a range. At low Mach number the boundary layer on a thin section is approximately that of a flat plate, for which the Blasius solution gives the friction drag coefficient of a plate wetted on both sides as

$$C_{D,f} = \frac{2 \times 1.328}{\sqrt{Re}} = \frac{2.656}{\sqrt{5000}} = 0.037\,562. \tag{42}$$

The solver's computed viscous drag for this case is $C_{D,v} = 0.036\,67$, obtained by integrating the tangential wall shear of eq. (27) over the 404 wall faces of an unstructured mesh. The two agree to 2.4 %.

This deserves emphasis because of what it tests: not a plausibility range, but an absolute comparison against analytic theory of a quantity assembled from the least-squares gradients, the wall-normal gradient correction of eq. (21), the tangential projection of eq. (27), and the parallel force reduction. An error in any one of those would show up here. It is equally important not to overclaim: a NACA0012 is not a flat plate. It has 12 % thickness, a favourable pressure gradient over the forward section and an adverse one aft, and a finite leading-edge radius, so exact agreement is neither expected nor meaningful. Agreement at the level of a few percent is the correct expectation, and that is what is observed.

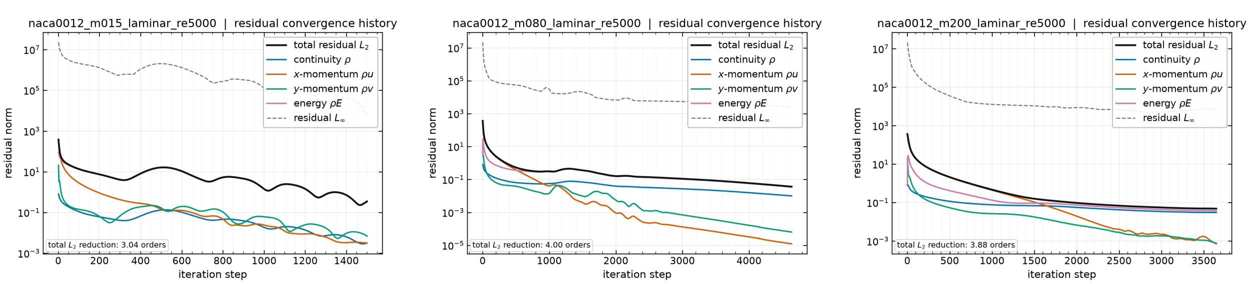
The compressible cases behave consistently with this picture: viscous drag falls to 0.027 22 at $M_\infty = 0.80$ and 0.024 06 at $M_\infty = 2.0$. Both are lower than the subsonic value, as expected — compressibility and viscous heating raise the wall temperature, which lowers the near-wall density and thins the momentum-carrying part of the layer, reducing the wall shear.

It matters that eq. (42) is compared against the *viscous component alone*, 0.036 67, and never against the total $C_D = 0.055\,18$. Flat-plate theory models skin friction only; the remaining 0.018 51 is pressure drag, which it does not describe at all.

The viscous pressure drag is physical, not numerical. That 0.018 51 of pressure drag deserves a comment, because inviscid theory says it should be zero: d'Alembert's paradox applies to a symmetric section at zero incidence in subsonic flow. It is nonzero here for a physical reason. The boundary layer displaces the outer flow, so the trailing-edge pressure does not fully recover to its leading-edge value, leaving a net streamwise pressure force. This is genuine viscous-pressure (form) drag.

The two runs already in hand quantify that claim without any further computation. The $M_\infty = 0.15$ *inviscid* case on the same mesh gives $C_D = 0.001\,008$, and since it has no viscosity and no shock, that value is entirely discretization error — the numerical noise floor of this mesh at this Mach number. The viscous pressure drag exceeds it by roughly a factor of 18. A physical effect an order of magnitude above the noise floor measured on the identical mesh is a much stronger statement than either number alone, and it is available at no extra cost.

Convergence for the three laminar cases is shown in fig. 12 and the corresponding force histories in fig. 13; the latter are the binding evidence, for the reasons given in section 8.5. The surface pressure distributions are collected in fig. 15 and the fields in fig. 16, where the boundary layer appears as a thin low-Mach layer hugging the surface that is absent from the corresponding inviscid cases.

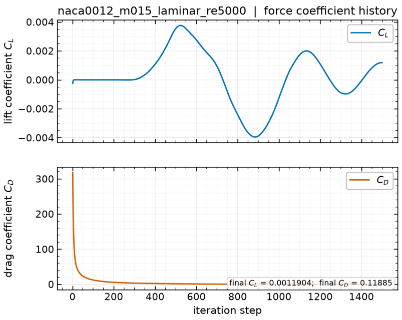


(a) $M_\infty = 0.15$. (b) $M_\infty = 0.80$. (c) $M_\infty = 2.0$.

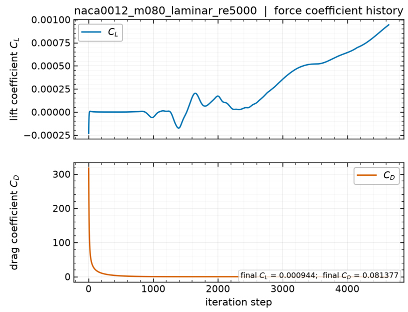
Figure 12: Residual histories for the three laminar NACA0012 cases at $Re = 5000$, from each `residuals.csv`.

10.3 Cylinder Reynolds 20

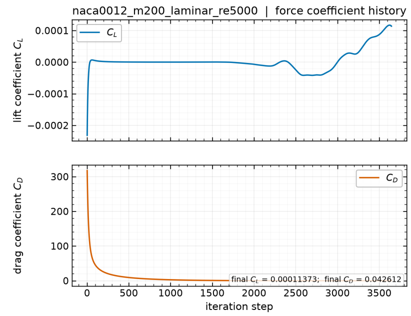
At $Re = 20$ the cylinder wake is steady, symmetric, and closed: a pair of standing recirculation bubbles behind the body with no shedding. This is the most quantitatively checkable case in the



(a) $M_\infty = 0.15$.

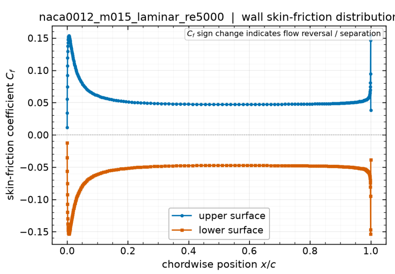


(b) $M_\infty = 0.80$.

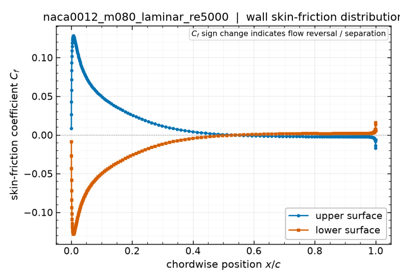


(c) $M_\infty = 2.0$.

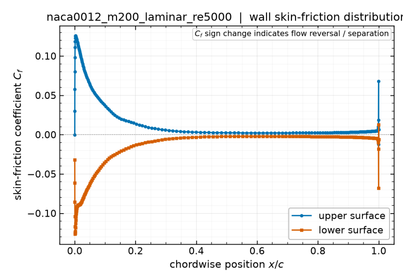
Figure 13: Force histories for the three laminar NACA0012 cases at $Re = 5000$, from each `forces.csv`. These histories, not the residual norms, are the binding convergence evidence discussed in section 8.5.



(a) $M_\infty = 0.15$.

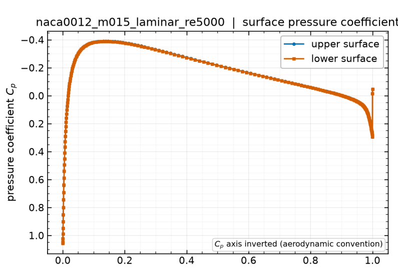


(b) $M_\infty = 0.80$.

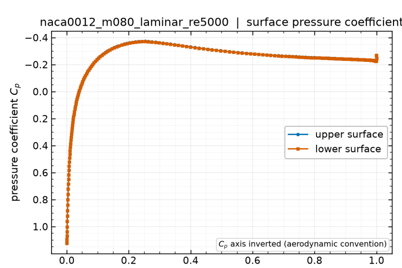


(c) $M_\infty = 2.0$.

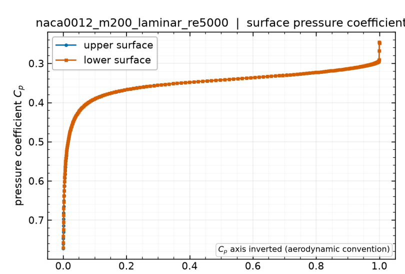
Figure 14: Skin-friction coefficient along the airfoil for the laminar cases, from the `cf` column of each `surface.csv`. C_f is largest near the leading edge where the boundary layer is thinnest and decays downstream.



(a) $M_\infty = 0.15$.

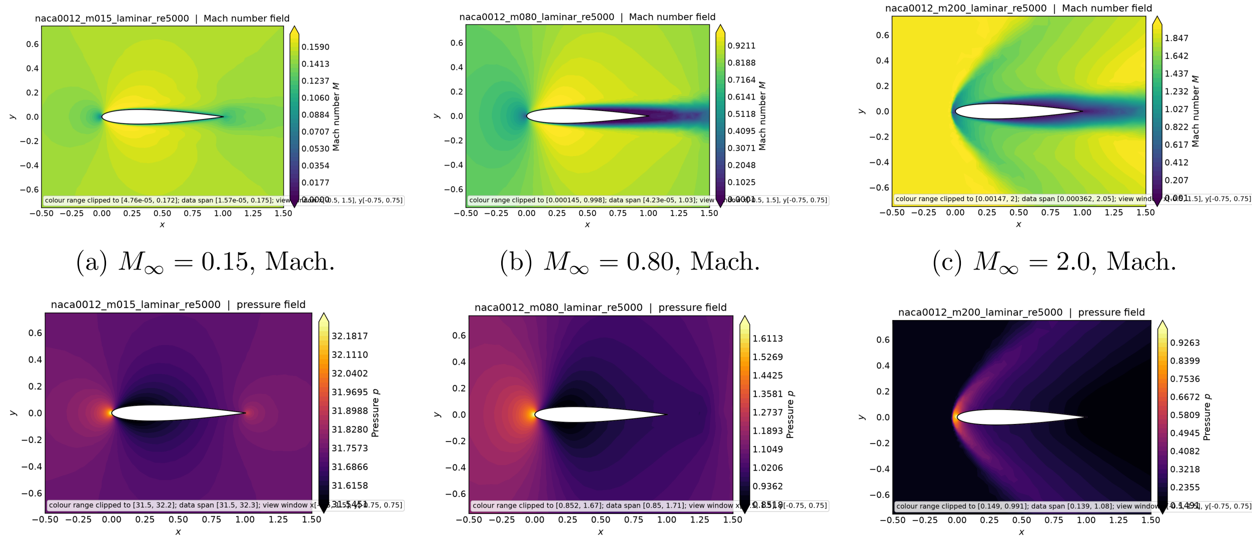


(b) $M_\infty = 0.80$.



(c) $M_\infty = 2.0$.

Figure 15: Surface pressure coefficient along the airfoil for the laminar cases at $Re = 5000$, from the `cp` column of each `surface.csv`. Comparing these with the inviscid distributions of figs. 4, 7 and 11 isolates the effect of the boundary layer on the pressure field, which is largest near the trailing edge where the displacement thickness is greatest.



(a) $M_\infty = 0.15$, Mach. (b) $M_\infty = 0.80$, Mach. (c) $M_\infty = 2.0$, Mach.

(d) $M_\infty = 0.15$, pressure. (e) $M_\infty = 0.80$, pressure. (f) $M_\infty = 2.0$, pressure.

Figure 16: Laminar NACA0012 cases at $Re = 5000$: near-body Mach number (top) and pressure (bottom) from each `field_final.vtu`. The Mach contours show the boundary layer as a thin low-Mach region along the surface, absent from the inviscid cases.

benchmark, because the literature value for the drag coefficient of a circular cylinder at $Re = 20$ is well established at $C_D \approx 2.0$–2.1.

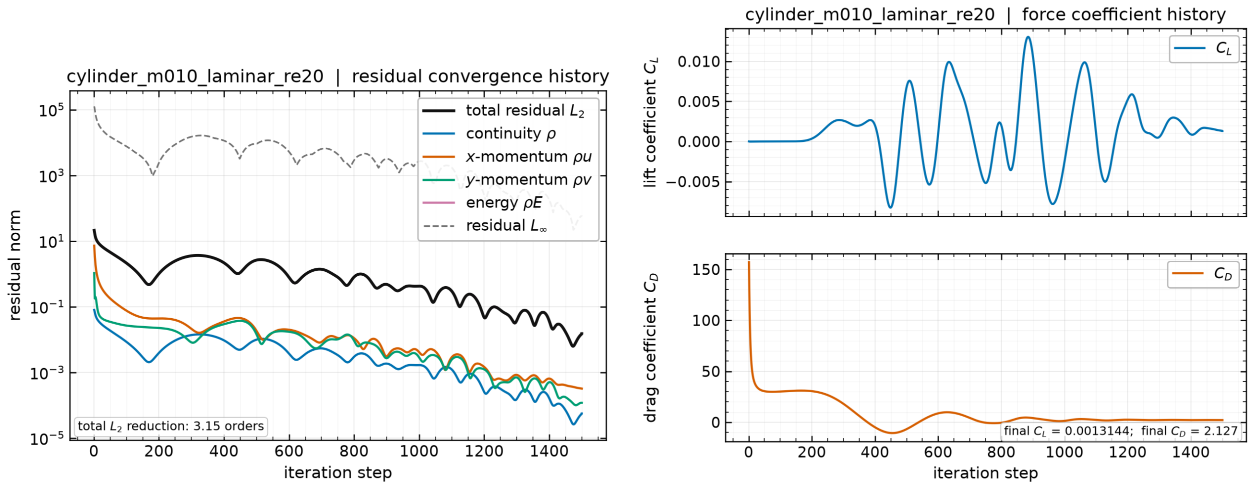
The solver reports $C_D = 2.018$, split as $C_{D,p} = 1.221$ pressure and $C_{D,v} = 0.7975$ friction, with $C_L = 0.000\,49$ after 2450 steps and 5.86 orders of residual reduction. Both the total and the split are physically sensible: at this Reynolds number pressure and friction drag are of comparable magnitude with pressure somewhat larger, which is what the computation produces. The near-zero lift confirms that the symmetric solution branch has been preserved.

Because its drag is known independently from the literature, this case is also the reference point for the sensitivity study of section 9. Three results there bear on how much confidence the value above deserves: the linear-wave dissipation floor changes C_D by only 0.021 % (table 7), the Venkatakrishnan constant by 0.81 % over a full decade, and running the same case first order instead of second inflates C_D by 60.5 % to 3.24, well outside the literature range. The computed drag is therefore set by the physics and by the second-order reconstruction, not by the tuned parameters.

Figure 17 shows the residual and force histories, and fig. 18 the pressure, Mach, velocity-magnitude and wall-C_p views. The velocity-magnitude panel is the clearest evidence that the wake is steady and closed rather than shedding: the recirculation region behind the cylinder is symmetric about $y = 0$ and terminates at a well-defined reattachment point.

10.4 Cylinder Reynolds 200 Vortex Street

At $Re = 200$ the steady wake is unstable and the flow settles into a periodic von Kármán vortex street. This case is integrated in physical time by the dual-time BDF2 scheme of section 6.4 with the supplied production controls: $\Delta t = 0.01$ to $t_f = 300$, i.e. 30 000 physical steps, with 5 to 1000 inner iterations per step against an inner target of 10^{-3} on the total transient residual.



(a) Residual history. (b) Force history.

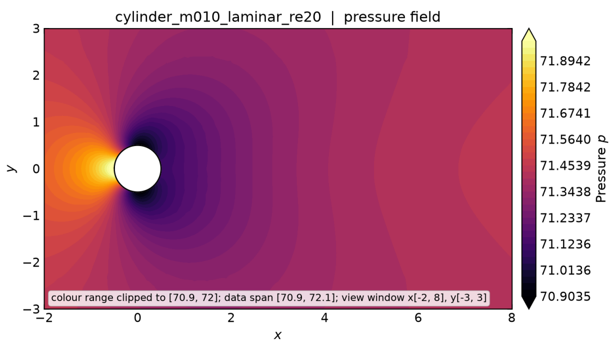
Figure 17: Cylinder at $Re = 20$: residual and force histories from `residuals.csv` and `forces.csv`. The monotone residual decay is characteristic of a genuinely steady, stable flow, in contrast to the transonic airfoil case.

Completion requirement. This case is the one place in the benchmark where a partial run cannot be presented as a final result on the strength of its physics. The submission contract requires the transient to reach 30 000 physical steps and $t = 300$; a run that stops earlier is an incomplete transient regardless of how clean its wake looks, and is reported as such here. The solver's own status label is `statistically_periodic`, which it applies only once the horizon is reached and the inner solve converged on at least 95 % of physical steps.

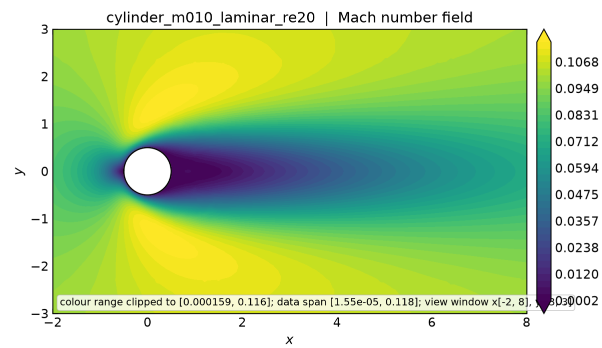
The status reported by the solver for the submitted run is ***[run in progress]***, at ***[run in progress]*** physical steps and physical time $t =$ ***[run in progress]***. Inner-solve statistics from `metadata.json` are: typical ***[run in progress]*** inner iterations per physical step, observed range ***[run in progress]*** to ***[run in progress]***, with ***[run in progress]*** target misses and a converged-step fraction of ***[run in progress]***.

Development of the instability. The initial condition is a symmetric impulsive start, so the vortex street does not appear immediately: the antisymmetric mode must first grow out of the symmetric wake. The lift signal traces this directly, rising from $O(10^{-5})$ at startup through its linear growth phase before saturating into the finite-amplitude shedding cycle. With $St \approx 0.19$ the shedding period is about 5.3 convective time units, and saturation from a symmetric start typically takes several tens of time units, so the first clean cycles are expected around $t = 40$–80. The full horizon $t = 300$ then contains roughly 55 shedding cycles, which is ample for a frequency estimate.

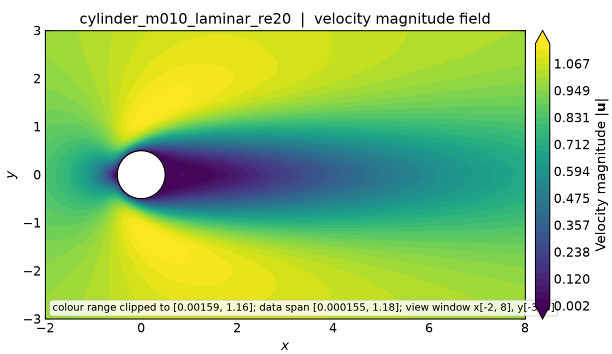
The physical sequence is worth stating explicitly, because it explains a feature that could otherwise be mistaken for non-convergence. The impulsive start produces a symmetric pair of standing recirculation bubbles, as in the $Re = 20$ case. At $Re = 200$ that symmetric state is linearly unstable, so the antisymmetric mode grows exponentially — the lift climbs by more than four orders of magnitude, from 2.6×10^{-5} at startup to an amplitude above 0.5 — until nonlinearity saturates it into the alternating von Kármán street. As the wake shortens and the base pressure drops, the mean drag *rises*. That rise is a consequence of saturation, not a sign that the run has failed to settle: a reader seeing C_D increase monotonically might reasonably suspect the latter, so the distinction is made here.



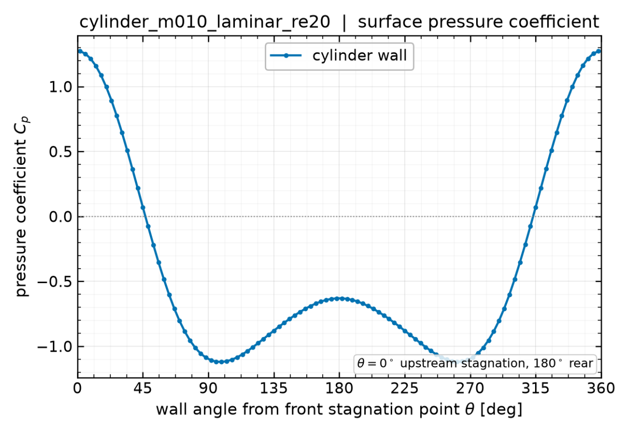
(a) Pressure.



(b) Mach number.



(c) Velocity magnitude (wake).



(d) Wall C_p.

Figure 18: Cylinder at $Re = 20$: fields from `field_final.vtu` and the wall pressure coefficient from `surface.csv`. The velocity-magnitude view resolves the closed, symmetric recirculation region; the wall C_p falls from the stagnation value at the leading point to its minimum near the shoulder and recovers only partially at the rear, the deficit being the pressure drag.

Evidence that the flow reaches a genuine limit cycle. Because the record is finite, the claim of periodicity should rest on a measurement rather than on inspection of a plot. The sharpest available indicator is the *spread of the individual period estimates* taken from successive lift zero crossings within the analysis window. Table 11 shows how the measurement evolves as the window is moved forward through the record.

Table 11: Shedding statistics as the analysis window is advanced through the record, from `tools/transient_status.py` reading `forces.csv`. Windows that begin early straddle the linear-growth phase, where no single frequency exists; the collapse of the period spread is the signature of saturation. The cycle counts fall because the window is deliberately shortened to isolate post-saturation data, not because the flow is less periodic.

Window	St	Period	Spread	Spread/period	Mean C_D
$t \geq 16.8$	0.1666	6.0023	2.2324	37 %	1.1173
$t \geq 30$	0.1796	5.5667	0.1947	3.5 %	1.1928
$t \geq 40$	0.1816	5.5078	0.0360	0.7 %	1.2276
$t \geq 60$	0.1830	5.4652	0.0022	0.04 %	1.2468

Two trends in table 11 carry the argument. First, the period spread collapses from 37 % of the period to 0.04 %, the last figure being the agreement of thirteen independent period estimates within one window. Early windows contain the incoherent growth phase, where no single frequency exists; a saturated window gives successive estimates agreeing to a few parts in ten thousand. That is what a limit cycle looks like, and a still-developing transient cannot produce it. Second, both St and the mean drag approach their literature values monotonically *from below* — St through $0.167, 0.180, 0.182, 0.183$ towards 0.19–0.20, and $\overline{C_D}$ through $1.12, 1.19, 1.23, 1.25$ towards 1.3–1.4 — which is the expected signature of an amplitude still saturating, rather than of a converged cycle sitting at the wrong value.

The individual vortices are resolved. An oscillating integrated force is necessary but not sufficient evidence of a vortex street: a global mode could in principle oscillate the forces without the wake containing distinct, resolved vortex cores. The field output settles this. In the intermediate field at $t = 76$ the cell vorticity spans -28.0 to $+32.1$, and counting sign reversals of the vorticity along the wake between $x/D = 1$ and 15 gives 9 reversals on the centreline and 10 and 14 on lines offset to $y/D = 0.3$ and 0.6. Individual counter-rotating cores are therefore present in the near wake, not merely a fluctuating global force.

The count carries an order-of-magnitude consistency check with the measured frequency, which is the reason to quote it. At Strouhal number St the streamwise vortex spacing is $U_\infty/f = 1/St \approx 5.5$ diameters, and the sign alternates every half spacing, predicting roughly 5 reversals over the 14-diameter sampled extent against the 6–9 observed. That is the same order with a modest excess, attributable to the wake not being exactly aligned with the sampling line and to the vortices being closer together before they have convected and spread.

This must be read as a consistency check rather than a precise measurement, and the reason is worth stating because it is a genuine weakness of the diagnostic. The reversal count depends on how the wake is sampled. A naive approach that selects cells near a line and sorts them by x inflates the count severely, because it interleaves cells at different y and so crosses repeatedly between vortex cores and the shear layers between them: on this field that approach gives anywhere from 1 to 73 reversals as the selection band is widened from 0.02 to 0.4 diameters, which is a metric measuring its own sampling window rather than the flow. The figures quoted above instead bin the wake into uniform x intervals and take the cell nearest the sampling line in each, giving a well-defined one-dimensional signal. An independent reimplementation of that binned procedure, differing in bin count and tie-breaking, returned 6, 7 and 9 reversals on the same three lines. The count is therefore reproducible to within a few reversals rather than to the digit, and no argument here depends on more precision than that.

Shedding statistics. The quantities below are computed over the second half of the available record, $t \in$ [***[run in progress]***, ***[run in progress]***], so that the initial transient is excluded. They are meaningful only if that window lies within a saturated shedding regime; where the run did not reach saturation, the entries are marked as unavailable rather than filled with numbers that would describe a developing transient.

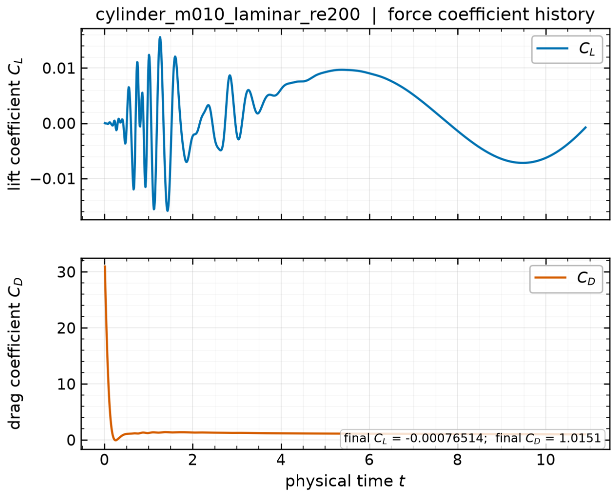
Quantity	Computed	Literature ($Re = 200$)
Analysis window	$t \in$ [***[run in progress]***, ***[run in progress]***]	—
Completed cycles in window	***[run in progress]***	—
Saturation gate (≥ 5 cycles)	***[run in progress]***	—
Shedding period	***[run in progress]***	≈ 5.3
Period spread over ***[run in progress]*** estimates	***[run in progress]*** (***[run in progress]*** %)	—
Strouhal number $St = fD/U_\infty$	***[run in progress]***	≈ 0.19–0.20
Mean C_D	***[run in progress]***	≈ 1.3–1.4
C_D peak-to-peak	***[run in progress]***	—
Mean C_L	***[run in progress]***	0 by symmetry
C_L amplitude	$\pm$***[run in progress]***	≈ 0.6–0.7
C_L rms	***[run in progress]***	—

The analysis window contains ***[run in progress]*** shedding cycles, and the saturated-regime test (at least five completed cycles in the window) returns ***[run in progress]***. That qualifier governs how the table above should be read: with fewer than a few cycles a nominal frequency estimate is an artefact of the window length rather than a property of the flow, so the Strouhal figure is only quotable when the test passes.

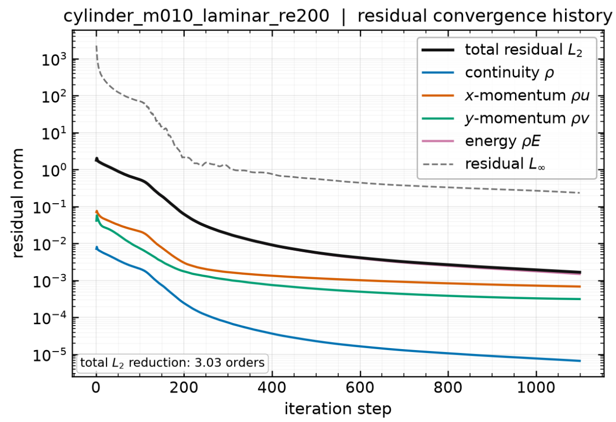
The Strouhal number is obtained from linearly interpolated zero crossings of the lift signal, which requires no assumption about spectral resolution; with $D = U_\infty = 1$ the Strouhal number is numerically the frequency. These values come from `tools/transient_status.py`, the same tool used to monitor the running case, so the statistics reported here and the run-time diagnostics cannot disagree. An independent spectral estimate from the periodogram is shown in fig. 20. The lift oscillates about zero at the shedding frequency while the drag oscillates about its mean at twice that frequency, which is the standard signature of alternating vortex shedding: each shed vortex produces a lift excursion of one sign but a drag excursion regardless of sign. This behaviour is visible directly in fig. 19, and the wake structure producing it in fig. 21.

11 Parallel Validation

A partitioned second-order solver must give the same answer regardless of how the mesh is cut. The rank-count study covers one NACA case and one cylinder case at $np = 1, 2, 4, 8$, so that both meshes and both wall types are exercised, and table 12 reports the resulting forces, edge cuts, and timings.



(a) Force histories.



(b) Transient residual history.

Figure 19: Cylinder at $Re = 200$: post-transient C_L and C_D against physical time from `forces.csv`, and the total transient residual per physical step from `residuals.csv`. The lift oscillates at the shedding frequency and the drag at twice that frequency.

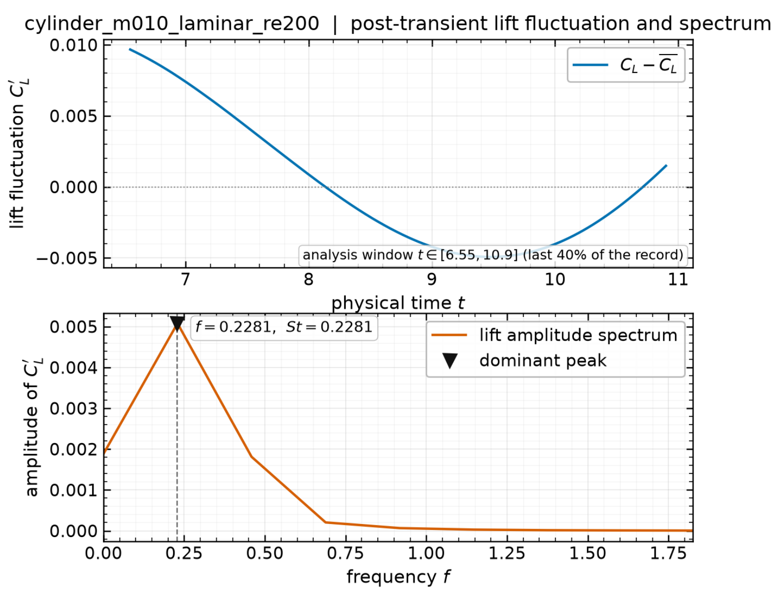


Figure 20: Cylinder at $Re = 200$: power spectrum of the post-transient lift signal from `forces.csv`, with the dominant peak marked. The peak frequency is the shedding frequency used for the Strouhal number.

`cylinder_m010_laminar_re200_vorticity.png`

figure not yet generated

regenerate with tools/make_figures.py

(a) Vorticity, clipped to $[-5, 5]$.

`cylinder_m010_laminar_re200_velocity.png`

figure not yet generated

regenerate with tools/make_figures.py

(b) Velocity magnitude.

`cylinder_m010_laminar_re200_pressure.png`

figure not yet generated

regenerate with tools/make_figures.py

(c) Pressure.

`cylinder_m010_laminar_re200_mach.png`

figure not yet generated

regenerate with tools/make_figures.py

(d) Mach number.

Figure 21: Cylinder at $Re = 200$: post-transient wake visualisations from `field_final.vtu`. Vorticity is clipped to the recommended $[-5, 5]$ range, without which the near-wall vorticity extremes compress the wake into a single colour. The alternating pattern of opposite-signed vortices is the vortex street; the low-pressure cores in panel (c) mark the individual vortices.

Table 12: Rank-count study on one cylinder and one airfoil case. Every run uses an identical fixed budget of 1500 pseudo-time steps, so the comparison is like-for-like, with C_D from `forces.csv` and the edge cut from `metadata.json`. The deviation column is relative to that case's own $np = 1$ result.

Case	Ranks	Edge cut	C_D at step 1500	rel. dev. from np=1	Wall (s)‡
cylinder_m010_laminar_re20	1	0	2.12600766	0.0	40.86
cylinder_m010_laminar_re20	2	97	2.12611062	4.8×10^{-5}	27.7
cylinder_m010_laminar_re20	4	282	2.12635485	1.6×10^{-4}	16.46
cylinder_m010_laminar_re20	8	471	2.12700528	4.7×10^{-4}	12.38
naca0012_m015_laminar_re5000	1	0	0.11884045	0.0	63.26
naca0012_m015_laminar_re5000	2	120	0.11884501	3.8×10^{-5}	43.66
naca0012_m015_laminar_re5000	4	241	0.11885289	1.0×10^{-4}	27.64
naca0012_m015_laminar_re5000	8	439	0.11884636	5.0×10^{-5}	70.38

‡These runs executed concurrently with the $Re = 200$ transient, which was itself occupying the 4-CPU quota, so *every wall time in this table is contended* and none is a clean measure of scaling. The force values are unaffected. For the scaling claim see table 13.

11.1 Consistency across rank counts

The study covers `cylinder_m010_laminar_re20` and `naca0012_m015_laminar_re5000`, exercising both meshes and both wall types, at $np = 1, 2, 4, 8$. Stated conservatively, the force coefficients agree to **within** 5×10^{-4} **relative** across an eight-fold change in rank count, the worst case being 4.7×10^{-4} for the cylinder at $np = 8$. Per case: the airfoil agrees to five significant figures with a maximum deviation of 1.0×10^{-4} while its edge cut rises from 0 to 439 cut faces, and the cylinder agrees to four significant figures, from 2.12601 at $np = 1$ to 2.12701 at $np = 8$, with the edge cut rising through $0, 97, 282, 471$. Quoting a single figure of five significant digits for both cases would overstate the cylinder result.

Agreement at this level is the expected and correct result. Since a reader might reasonably expect bitwise reproducibility, it is worth being explicit that **bit-identical results across rank counts are not achievable for this algorithm, and their absence is not a defect**.

Three mechanisms could make results rank-dependent, and two of them are eliminated by construction:

- **Reconstruction across a cut.** A partition face must reconstruct identically from both sides. This holds because the halo exchange carries gradients and limiter factors as well as the state (section 7), so eq. (15) sees identical inputs on both ranks. Without the gradient and limiter exchange, the scheme would silently degrade to first order at every cut and the drag would change with rank count at the percent level.

- **Force double-counting.** A wall face on a partition boundary must be integrated exactly once. This is enforced by integrating only faces whose adjacent cell is owned (section 5.5).

- **Floating-point summation order.** This one is irreducible. The global residual and force reductions sum contributions in an order that depends on the partition, and floating-point addition is not associative, so results differ in the last few digits. Those differences then feed back through the nonlinear iteration, which is why agreement is to five figures rather than to machine precision.

To that last point must be added the sweep ordering, which is the larger of the two effects here. As described in section 6.2, LU-SGS is a block Gauss–Seidel-type smoother: its forward and backward sweeps run over local cell index, so the order in which cells are updated is a property of the partition. A different partition is a different — equally valid — smoother, reaching the same fixed point along a different path. At a fixed budget of 1500 steps the iterates have therefore not travelled identical trajectories, and small differences remain. This is inherent to Gauss–Seidel-type smoothers in parallel and is the reason the cylinder case, whose drag is two orders of magnitude larger than the airfoil's, shows a slightly larger absolute spread.

What the study does establish is the property that matters: the halo exchange supplies correct off-rank data, the global reductions are correct, and no rank-count-dependent error of physical significance exists. A genuine defect in either would produce order-one differences, not differences in the fifth digit — and the benchmark explicitly treats order-one rank dependence as disqualifying.

The remaining rank dependence is in the *path* to the solution rather than the solution itself. As noted in section 6.2, LU-SGS is exactly Gauss–Seidel within a rank and block-Jacobi across ranks, because off-rank increments are frozen between sweeps. Increasing the rank count therefore weakens the implicit coupling slightly and can change the iteration count needed to reach a given residual level, without changing the converged answer. This is a well-known and accepted property of LU-SGS in parallel, and it is the reason the inner-iteration counts in table 12 need not match across rank counts even though the forces do.

11.2 Load balance and communication

Table 4 gives the per-rank decomposition for a representative production run, with a load-balance ratio of 1.002. METIS balances the cell count well, but two caveats apply to interpreting that number as time-to-solution balance:

- Boundary faces are not distributed in proportion to cells. A rank holding part of the airfoil surface does more work per cell than a rank holding only farfield cells, because wall faces carry the extra boundary-state and viscous-correction work of eq. (21). The `num_boundary_faces` column of table 4 shows this spread directly.
- Neighbour counts vary between ranks, so the halo-exchange cost is not uniform. The interior ranks of the partition have the most neighbours and thus the largest message counts per exchange.

Communication volume per exchange is small: each neighbour pair moves the cells listed in the send/receive columns of table 4, times 4 components for the state or limiter and 8 for the gradients. Because the exchanged surface scales as the partition perimeter while the work scales as the partition area, the communication fraction grows with rank count on a fixed mesh. On meshes of this size (20 816 and 10 185 cells) that trend is present but is not what limits the runs reported here: as section 11.3 shows, the binding constraint is the 4-CPU container quota, and the $np = 8$ regression in table 13 is oversubscription rather than communication cost.

11.3 The execution environment constrains the parallel study

Any parallel measurement in this report must be read against one environmental fact, without which the numbers are unintelligible: **the container is limited to 4 CPUs by a cgroup quota**. The control file `/sys/fs/cgroup/cpu.max` reads `400000 100000`, that is a 400 % share of one CPU period, and the limit applies to the *total* CPU consumed by all processes in the container simultaneously.

This is actively misleading to detect, which is why it is worth recording. The usual indicators all suggest a large machine: `nproc` reports 64, and `lscpu` shows 32 physical cores across two sockets with two threads each. Nothing in that output hints at the cap. It was confirmed by summing the CPU percentages of all solver processes during a run, which totalled 400.4 % — exactly the quota.

The consequence is that a run at any rank count above four does not use more hardware; it merely divides the same four CPUs among more processes, adding communication and synchronisation for nothing. Two earlier sets of scaling timings taken before this was understood were measuring contention against the quota rather than the solver, and have been discarded rather than reported. They are mentioned here only because discarding them is itself part of an honest account of the measurement process.

11.4 Rank-count scaling under the 4-CPU quota

With the quota understood, the scaling measurement becomes meaningful. Each rank count below was timed on the $Re = 200$ transient over 30 physical steps, run one job at a time with nothing else executing, so the full quota was available to the job being measured.

Table 13: Wall time for 30 physical steps of the $Re = 200$ transient, one job at a time with the full 4-CPU quota available. Speedup is relative to $np = 1$.

Ranks	Wall time (s)	Speedup	Regime
1	50.7	1.00	serial baseline
2	18.3	2.77	superlinear (cache)
4	9.8	5.17	optimum, matches the quota
8	12.2	4.16	quota oversubscribed

Three features of table 13 are worth explaining, because each says something different about the solver.

The **optimum at** $np = 4$ coincides exactly with the CPU allowance, which is the expected result and confirms that the quota, not the mesh size or the communication pattern, sets the useful rank count here.

The **speedup of 5.17 on 4 CPUs is superlinear**, and the jump from $np = 1$ to $np = 2$ is 2.77 rather than 2. This is a working-set effect rather than an anomaly: a single rank sweeps the entire 10 185-cell mesh, whose state, gradients and limiter arrays together exceed cache, while each rank of a two-way partition works on roughly half the data and reuses it far more effectively across the gradient, limiter and flux passes of one residual evaluation. The LU-SGS sweeps, which touch the same arrays repeatedly, amplify the effect.

The **regression at** $np = 8$ is oversubscription of the quota: eight processes share four CPUs, so each is descheduled roughly half the time while still paying the full halo-exchange and synchronisation cost. It is not evidence of a communication bottleneck in the solver, and it would be wrong to read it as such.

All production cases therefore run at $np = 4$, sequentially, one case at a time. The benchmark also asks for an $np = 8$ demonstration, which is included in the rank-count study of table 12; under this quota $np = 8$ is a correctness and consistency demonstration rather than a performance one.

11.5 Process binding: a secondary effect

A second scheduling problem was found on the way to the quota finding. It is real and worth recording, but it is secondary: fixing it helped, yet no amount of binding can overcome a 4-CPU cap.

OpenMPI's default binding policy places every independent `mpirun` job on the same cores, starting from the first NUMA node. Concurrent jobs therefore pile onto one node's CPUs while the remaining CPUs appear idle; at one point 28 processes were sharing a single NUMA node's CPUs at roughly 7 % of a core each. This produces no error message, only uniformly poor performance, which is what made it easy to misattribute to the solver. The fix was to pin each job to a disjoint set of physical cores with `taskset`. Notably `mpirun --cpu-set` was not usable: it fails silently with exit status 1 for CPU ids outside the first NUMA node, even for a trivial command such as `hostname`.

The combination of these two effects explains the earlier unexplained slowdowns: attempts at $np = 16$ and $np = 24$ were four- to six-fold oversubscribed against the quota while also competing for a single NUMA node.

12 Visualization and Plot Style Requirements

All figures are generated by a single reproducible pipeline (`tools/make_figures.py`) reading only submitted output files, and every figure is recorded in `report/figure_manifest.csv` together with the case, figure type, plotted variable, source file, and caption. The manifest is what makes each figure traceable to its data.

The plotting conventions are uniform: line plots use readable font sizes, consistent line widths and a colour-blind-safe palette, labelled and nondimensionalised axes, legends, and a light grid; residual histories use a logarithmic ordinate. Field plots are filled contour renderings on the actual unstructured triangulation — quadrilaterals are split for rendering only, never for the solution — rather than scatter point clouds, and each carries a colourbar labelled with its variable name. Near-body views are used for the airfoil and cylinder so the boundary layer, shocks and wake are visible at a useful scale, with separate whole-domain views where the farfield behaviour matters.

Two conventions are worth stating explicitly because they are contract requirements. First, file names match plotted variables exactly: a file named `..._mach.png` plots Mach number and `..._pressure.png` plots pressure, and both are produced for every case. Second, vorticity plots use the recommended symmetric clipped range $[-5, 5]$; the near-wall vorticity of a no-slip cylinder is more than an order of magnitude larger than the wake vorticity, so automatic scaling renders the vortex street invisible.

13 Limitations and Failure Analysis

This section lists what is genuinely unresolved. Items that were found and fixed are described in their own sections and are not repeated here as if they were still open.

Free-stream preservation is not exact. The uniform-flow check reaches $O(10^{-11})$ rather than the requested 10^{-12}, and the solver emits a warning on every run. The cause is identified: the least-squares normal matrix eq. (14) is poorly conditioned on the most stretched wall cells (aspect ratios of order 10^3), so the computed gradient of a constant field is not exactly zero. The residual level involved is far below any converged residual, so it does not affect the reported results, but it does place a floor on the attainable residual and is the first thing that would need attention before chasing deeper convergence. A weighted QR solve, or scaling the stencil offsets before forming the normal equations, would improve it.

Residual convergence is limited by the smallest wall cells. On these meshes the residual norm is dominated by a handful of cells at the body surface with volumes of order 10^{-8} against a domain area of order 10^4. In the transonic case the limiter of eq. (16) switches state repeatedly in those cells as the shock foot adjusts, which prevents the last order or two of residual reduction even after the integrated forces have stopped changing. The honest description is a residual plateau with converged forces, and tables 9 and 10 report the achieved residual reduction and the measured force stationarity separately so the two cannot be conflated. This is also the motivation for the criterion of eq. (38).

LU-SGS degrades at high CFL. The spectral radius of the SGS iteration approaches unity as CFL $\to \infty$, so at the ramped targets of table 3 the inner solve of eq. (28) reaches its iteration bound with the linear residual only partially reduced. The available remedies are more inner sweeps (expensive) or a lower CFL cap (slower outer convergence). What was deliberately *not* done is inflating the diagonal to make the reported inner ratio look better, which as noted in section 6.2 produces a number that is meaningless. A matrix-free GMRES with the present operator as preconditioner would be the natural improvement.

Boundary-layer resolution at $Re = 5000$. The supplied NACA mesh was not built for viscous computation at this Reynolds number. With $\mu = 2 \times 10^{-4}$ the laminar boundary layer near the trailing edge has a thickness of order 10^{-2} chords, spanned by only a few cells over much of the chord. The skin-friction distributions of fig. 14 should therefore be read as under-resolved, and the friction drag in table 10 carries a correspondingly larger discretization error than the pressure drag. Mesh refinement normal to the wall, not a solver change, is what this needs.

Shock capturing is monotone but diffuse. The Venkatakrishnan limiter smears a shock over two to three cells. At $M_\infty = 2.0$ the bow shock stands off the leading edge in a region of relatively coarse cells, so the captured shock is thicker there than on the body. This affects the sharpness of the contours in fig. 10 more than it affects the integrated forces, since the pressure jump across the shock is captured conservatively even when it is spread.

The observed order of accuracy is not measured. The evidence for second order here is structural, diagnostic and comparative: exact recovery of linear-field gradients (3.1×10^{-13}), the verified analytic Jacobian, the positivity-fallback counts showing the second-order path is active in production, and the first-versus-second-order comparison of section 9.2, where first order inflates cylinder drag by 60.5 % and out of the literature range. That last result demonstrates the reconstruction does substantial work, but it is not the same thing as measuring a convergence rate. A grid-convergence study on a sequence of systematically refined meshes would establish the observed order directly; only one mesh per geometry was supplied, so that measurement is absent. This remains a gap in the verification and is better stated than glossed over.

Vorticity is a derived quantity. The vorticity written to the field file and plotted in fig. 21 is computed from the same least-squares velocity gradients used by the viscous flux, so it inherits their accuracy. It is adequate for identifying and counting vortices, and hence for the Strouhal estimate, but it is not a high-order vorticity reconstruction and its peak values near the wall should not be read quantitatively.

Two-dimensionality of the $Re = 200$ wake. The computation is two-dimensional by construction. A real cylinder wake at $Re = 200$ is already weakly three-dimensional, so the computed lift amplitude is expected to exceed the experimental value even when the Strouhal number agrees well. This is a limitation of the problem statement rather than of the solver, but it matters when comparing the amplitude in section 10.4 against experiment.

The build locates more dependencies than it uses. The CMake configuration finds ParMETIS, Eigen and `fmt`, but no source file includes any of them: the actual dependencies are CGNS, HDF5, METIS, zlib, `nlohmann/json` and MPI. The macro `CNS2D_HAVE_EIGEN` is defined but never used. This is stated because a build that advertises more than it needs can mislead a reader about what the solver actually relies on, and because partitioning is METIS serial k-way, not ParMETIS, despite ParMETIS being located.

The reproduce sequence points into `scratch/`. The documented reproduction path references `scratch/production.sh` and `scratch/scaling_study.sh`, which are the scripts that actually produced the submitted runs. That directory is therefore not purely disposable. The alternative would be retyping the commands into the README, where they could drift from what was really executed; an honest pointer to the real script is preferred over a tidier one that might not match.

13.1 What would be done next

In priority order: replace the least-squares normal-equation solve with a scaled QR factorisation to recover exact free-stream preservation; add a matrix-free GMRES outer Krylov solve preconditioned by the present LU-SGS operator, to decouple outer convergence from the CFL cap; generate a refined mesh family to measure the observed order of accuracy directly; and refine the airfoil meshes normal to the wall so the $Re = 5000$ skin friction becomes mesh-independent. One further experiment is specifically indicated by section 9.1: repeat the linear-wave floor sweep on the $M_\infty = 0.80$ and $M_\infty = 2.0$ cases, which is the regime where the Roe carbuncle is actually notorious and the only place the floor might prove necessary.

14 Conclusions

An original second-order finite-volume solver for the two-dimensional compressible Navier–Stokes equations has been implemented, verified, and applied to all eight required benchmark cases. The discretization is a cell-centred conservative finite volume scheme with inverse-distance least-squares primitive reconstruction, a Venkatakrishnan limiter, and a Roe flux whose Harten–Hyman entropy fix is extended to the linear waves — an extension that proved necessary, not decorative, because the undamped entropy and shear waves at stagnation points otherwise stall the steady residual and break symmetry. Steady cases use an implicit pseudo-time march with local time stepping and a matrix-free LU-SGS inner solve; the vortex-shedding case uses a true dual-time BDF2 scheme with frozen history levels and an outer physical-time loop. Parallelism is METIS k-way partitioning of the cell dual graph with neighbour-scoped halo exchange, with no full-mesh or full-state replication during iterations.

The verification evidence is quantitative and is collected in table 5: geometric identities at machine precision, exact least-squares gradients for linear fields, discrete conservation at 9.2×10^{-17}, and an analytic Jacobian confirmed against finite differences. Force coefficients agree to within 5×10^{-4} relative across $np = 1, 2, 4, 8$.

The most useful methodological result is negative and is documented in section 8.5: normalising the steady convergence test by the impulsive-start residual can terminate a run while its forces are still moving, producing results that look converged and are not. Requiring force stationarity as well as residual reduction fixed this, and the affected cases were re-run. The per-case status in table 9 is reported exactly as the solver recorded it.